INTRODUCTION

- *Metal powders* are the main constituent of a P/M component.
- The final properties of the finished P/M part depend on the powder characteristics such as size, shape, and surface area.
- For this reason, a single powder production method cannot be used for all the applications.
- Method of powder production is decided mainly by the final properties required in the component.
- Hence a variety of powder production methods have been in use to meet the diverse product requirements.

Major powder production methods

- ***Machining or milling:***
- Brittle materials like beryllium are made into powder form by machining a cast billet and reducing the swarfs by impact milling or by grinding to fine powders.
- This method, though suitable for brittle materials, has also been used for ductile materials like titanium alloys.
- Cast titanium alloys are converted into brittle hydrides by reacting them with hydrogen under vacuum and then milling them into powder.
- The hydrogen is subsequently driven off by dehydriding to restore the ductility.

- ***Preparation of powders* from its compounds** can be done through reduction of oxides by gaseous or solid reductants, electrolysis, thermal decomposition of compounds, reduction in molten state or by precipitation from aqueous solution.

- **Low-melting point metals** such as zinc and tin are made by deposition from vapour phase.

- **Atomization:**
- Disintegration of a molten stream of metal using high-velocity fluid jets.
- Fluids used include water and gases (nitrogen, argon).
- Water atomization is a very commonly used method for large scale production of powders.
- Gas atomization is used when oxidation of powders is to be avoided or when spherical powders are needed.
- Atomization has also other variations like Centrifugal Atomization, Rotating Electrode Process (REP), Plasma Rotating Electrode Process (PREP) as well as Soluble Gas Atomization.

Rapid solidification

Powders of metal and alloys can be produced by cooling them very rapidly (cooling rates > 105°C/s).
Techniques such as melt spinning, melt extraction fall under this catogory.
The products are usually obtained as ribbons or fibres which are subsequently crushed into powders and used.
Atomization techniques like rotating electrode process also fall in this category giving metal and alloy powders directly.
The main **advantage** of rapid solidification is its **ability to produce non-equilibrium structures** including amorphous metals (or metallic glasses), which have novel properties and hence potential applications in various fields.

POWDER PRODUCTION METHODS

- Broadly classified as **mechanical, chemical or physical methods** based on the nature of energy used for production.

- Mechanical Methods
- Cheapest employed for powder production.
- Mechanical methods involve the use of mechanical forces such as compressive force, shear or impact to effect particle size reduction of bulk materials.
- A number of methods are available to achieve size reduction.

Milling

- Milling of materials is of economic importance for P/M industry since mechanical comminution of **hard and brittle as well as soft and ductile materials** is the most widely used method of powder production.
- **Hammer and rod mills** are used to mill **spongy cakes of oxide-reduced, atomized or electrolytic powders**.
- Mechanical comminution has been successfully employed for milling **electrolytic iron, bismuth, beryllium, metal hydrides as well as chemically-embrittled materials like sensitized stainless steel.**

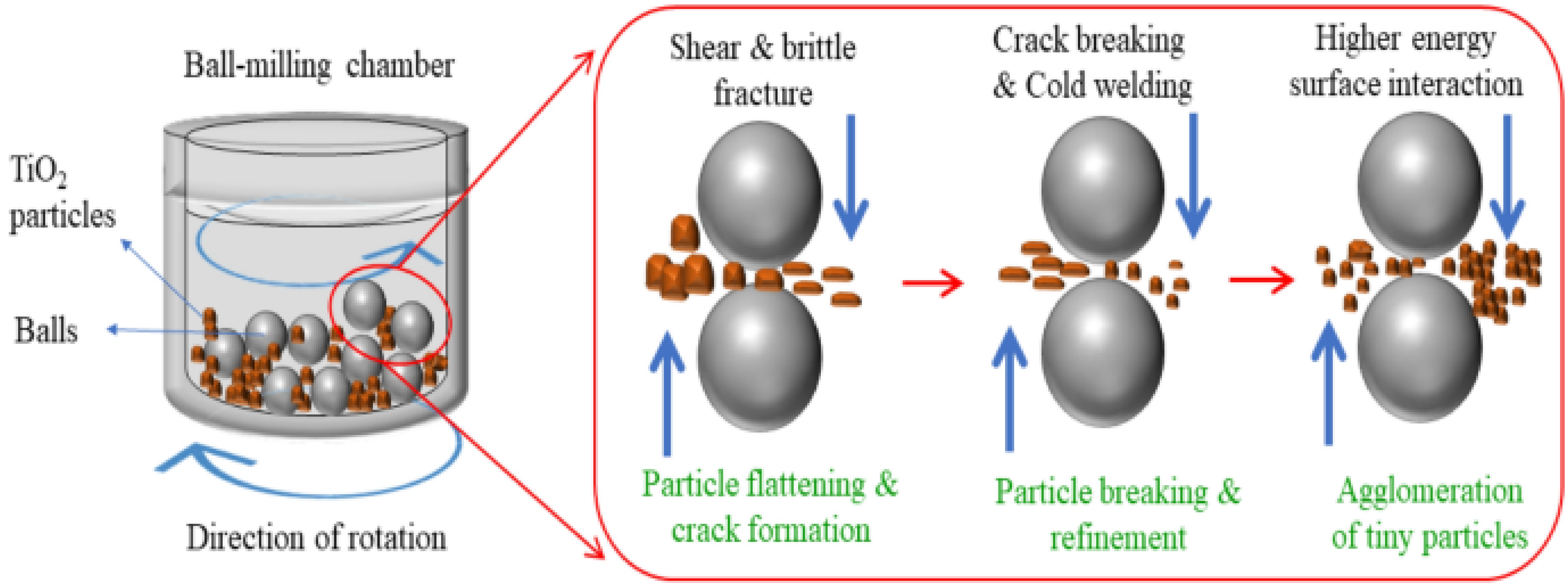
- During milling, the forces acting on the material being milled can be **impact, attrition, shear and compression**.
- While **impact is due to striking of one powder particle against another**,
- **Attrition refers to the production of wear debris due to the rubbing action between two particles**
- **Shear is due to cleaving or cutting of particles resulting in fracturing**
- **Compression is the application of compressive forces by crushing or squeezing of the particles usually associated with jaw crushers.**
- Due to these forces, deformation, cold welding and fracturing occur in hard as well as soft and ductile materials.

Main objectives of milling or grinding

(i) Particle size reduction (main purpose).
(ii) Particle size growth.
(iii) Shape change (to obtain flaky shape in aluminium for use in paints).
(iv) Agglomeration (putting the particles together).
(v) Solid state alloying (Mechanical Alloying).
(vi) Solid state mixing or blending two or more materials which are not easily mixed.
(vii) Modification of material properties, (e.g. altering the density or flowability of powders).

Mechanism of Milling

- Milling of coarse or bulk raw material **occurs in stages.**
- In the initial stage of milling of ductile materials, individual particles are deformed by micro forging resulting in welding and fracture.
- Eventually the particles become so severely cold worked that they fracture into fragments of smaller and smaller dimensions resulting in agglomerates due to coupling forces.
- After a period of cold welding and fracturing, steady-state equilibrium is achieved.
- Constant welding, fracturing and rewelding produce a composite structure containing fragmented composite particles.



Ball-milling chamber
TiO_2 particles
Balls
Direction of rotation
Shear & brittle fracture
Crack breaking & Cold welding
Higher energy surface interaction
Particle flattening & crack formation
Particle breaking & refinement
Agglomeration of tiny particles

Events during milling

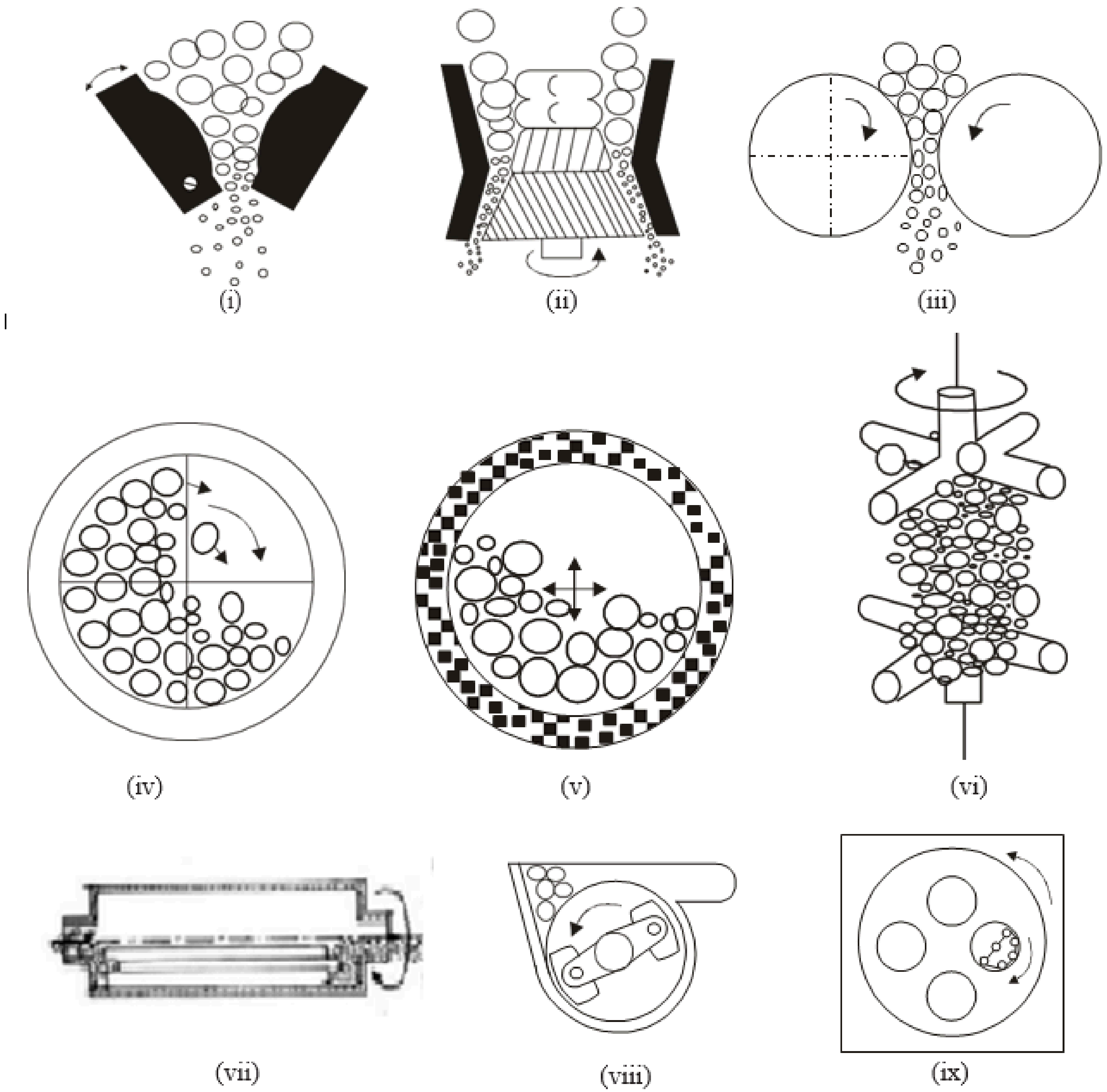
(i) Micro forging.
(ii) Fracture.
(iii) Agglomeration.
   i. Deagglomeration

Micro forging refers to the process during milling in which individual particles or a group of particles are impacted repeatedly so that they **flatten with very little change in mass**.

- After a period of milling, individual particles deform to such an extent that **cracks initiate and propagate resulting in fracture**.
- Mechanical interlocking due to atomic bonding or welding may lead to **agglomeration.**
- The process that **breaks up agglomerates** is known as **deagglomeration.**

- Milling of metal powders **results in external shape and internal structural** changes.
- The extent of such changes is determined by milling parameters, milling environment and the physical and chemical properties of the metal or alloy powder being milled.
- Mechanical mills have been successfully employed to achieve innovative P/M products not possible by conventional processing.

Milling Equipment



Comminution equipment

i. jaw crusher,
ii. gyratory crusher,
iii. roll crusher,
iv. ball mill,
(v) vibratory ball mill,
(vi) attritor,
(vii) rod mill,
(viii) hammer mill, and
(ix) planetary mill.

- A variety of equipment are available for comminution depending on the nature of the material being comminuted or on the final particle sizes required.
- These equipment are generally classified as **crushers and mills**.
- **Crushing** is one of the main methods used for ceramic materials such as oxides of metals.
- Crushing is also used as a preliminary step in the reduction of bulk materials into powders.

- **Grinding** is employed as a method of powder production **for reactive metals** such as titanium, zirconium, niobium and tantalum.
- These metals are converted into hydrides by heating in a vessel under hydrogen atmosphere to make them brittle and then subsequently crushed to give powders which are further treated.
- Brittle metals like beryllium are crushed directly while ductile metals like copper and aluminium are deformed considerably, strain hardened and subsequently reduced to small flaky particles.

- **Ball mills are** the most popular in industry and are very often used in Powder Metallurgy for making both powders and powder mixtures.
- A ball mill consists of a cylindrical vessel rotating horizontally along its axis.
- The length of the cylinder is generally more or less equal to the diameter - **Globe**
- The material to be milled is charged along with the **grinding media**.
- The grinding media may be made of **hardened steel or tungsten carbide or other ceramics like agate, porcelain, alumina and zirconia**.
- When the drum rotates, friction against the drum walls entrains the balls to a certain height, from where they freely fall or roll down and, in doing so, grind the material by impact and attrition.
- Above a critical number of revolutions Nc (rpm), centrifuging action takes place resulting in inefficient milling. Below a certain number of revolutions also inefficient milling results.

- The value of Nc is given by

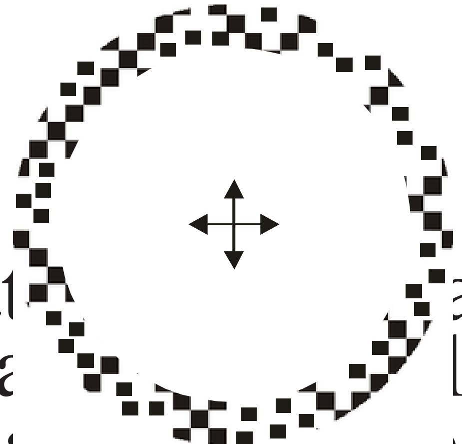
$$Nc = (60/2p)\ (2g/D)^{½}$$

- where D is the diameter of the mill in metres and g is the acceleration due to gravity.

- Milling can be either **dry milling or wet milling.**
- The mill is operated at 0.7–0.8 N_c for dry milling and at 0.5–0.65 N_c for wet milling.

- In the case of **dry milling**, about 25 vol% of powder is added along with about **1 wt% of a lubricant** such as stearic or oleic acid.
- For **wet milling**, 30–40 vol% of powder with **1 wt% of dispersing agent such as water, alcohol or hexane is employed.**
- Usually, the balls fill half the volume of the container.
- The ‘jars’ of the ball mills are covered with vulcanized rubber, polyurethane, alumina, porcelain, hard metal or stainless steel to reduce contamination.
- By using **‘autogenous milling**’—milling with **balls and linings made of the same material** as that being milled, contamination can be substantially minimized.

- Finer powders require longer periods of grinding.
- The need for intense grinding to achieve finer particle sizes has led to the design and development of **vibratory mills.**



- Comminution in vibrat[illegible] a process in which the mixture to be ground is subjected to vigorous a[illegible]ltitude of grinding bodies moving inside the container in different directions at varying speeds as a result of the applied vibratory loads.
- A vibratory mill consists of an electric motor connected to the shaft of the drum by an elastic coupling, the drum is usually lined on the inside with a wear resistant material.
- In operation, the container is filled up to **80% of its volume with the grinding bodies** and the starting material.

SCHEMATIC REPRESENTATION OF A VIBRATION MILL

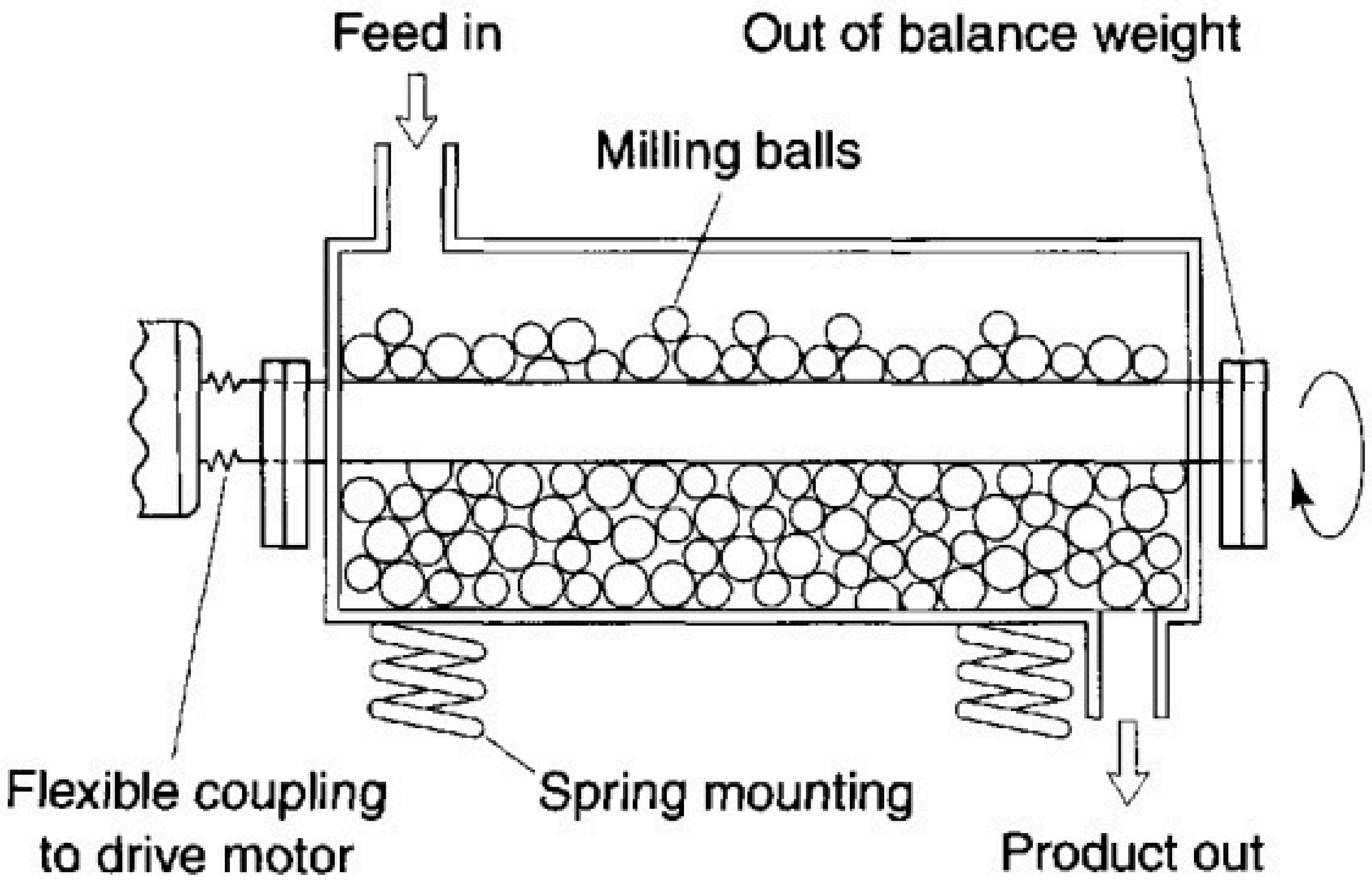
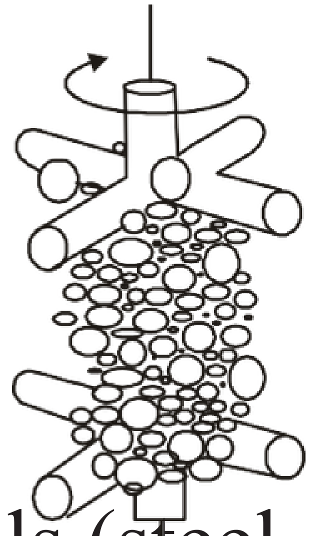


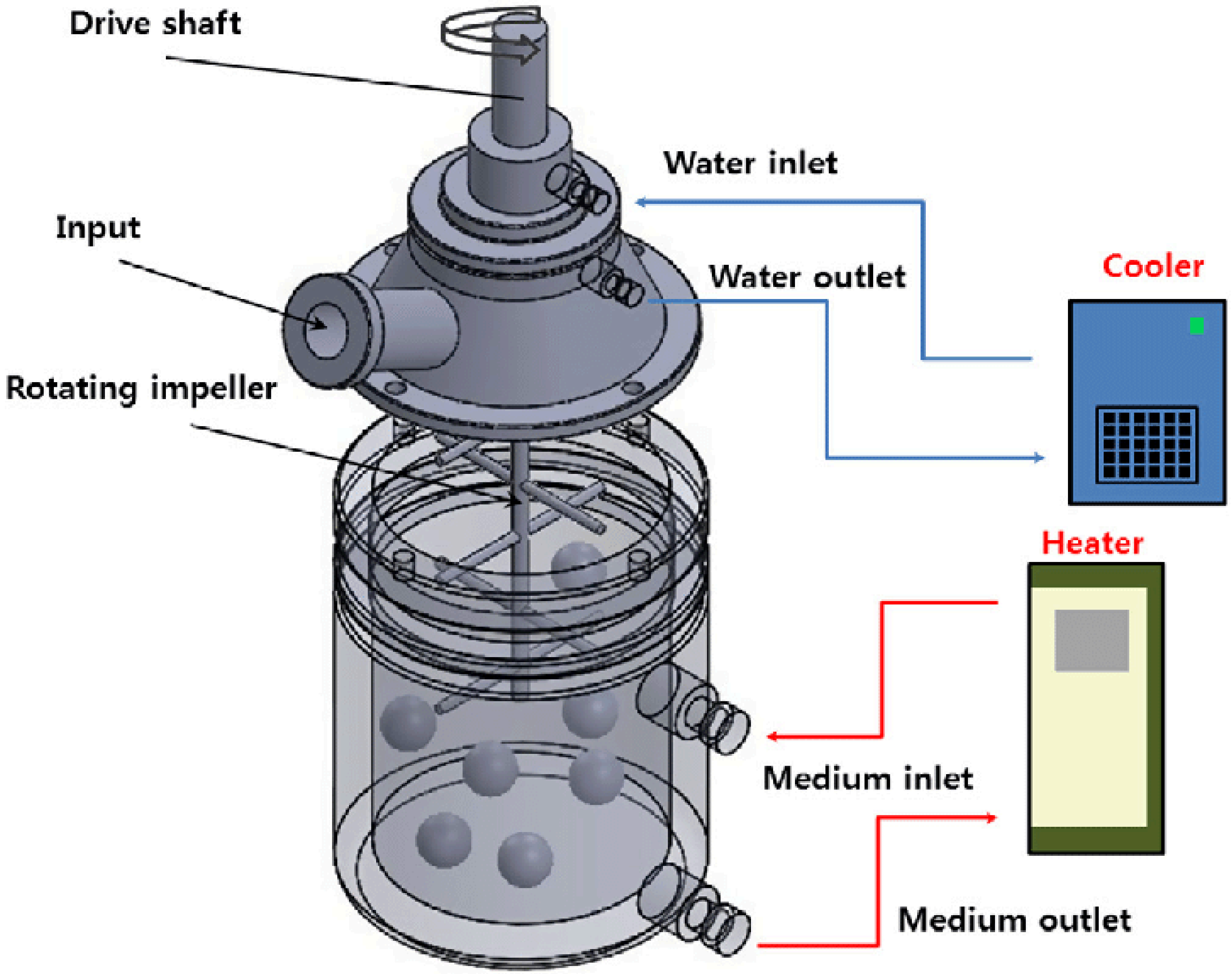
PHOTO CREDIT: AULTON'S PHARMACEUTICS: THE DESIGN AND MANUFACTURE OF MEDICINES

- **Attrition mill or attritor**, is another important type of mill in which the charge is ground to very fine sizes by the action of a vertical shaft with horizontal arms attached to it.



The powder along with the balls (steel, tungsten carbide or agate) is charged into the container in a certain ratio known as the ball-to-charge ratio, which may be 5:1, 10:1 or 15:1.

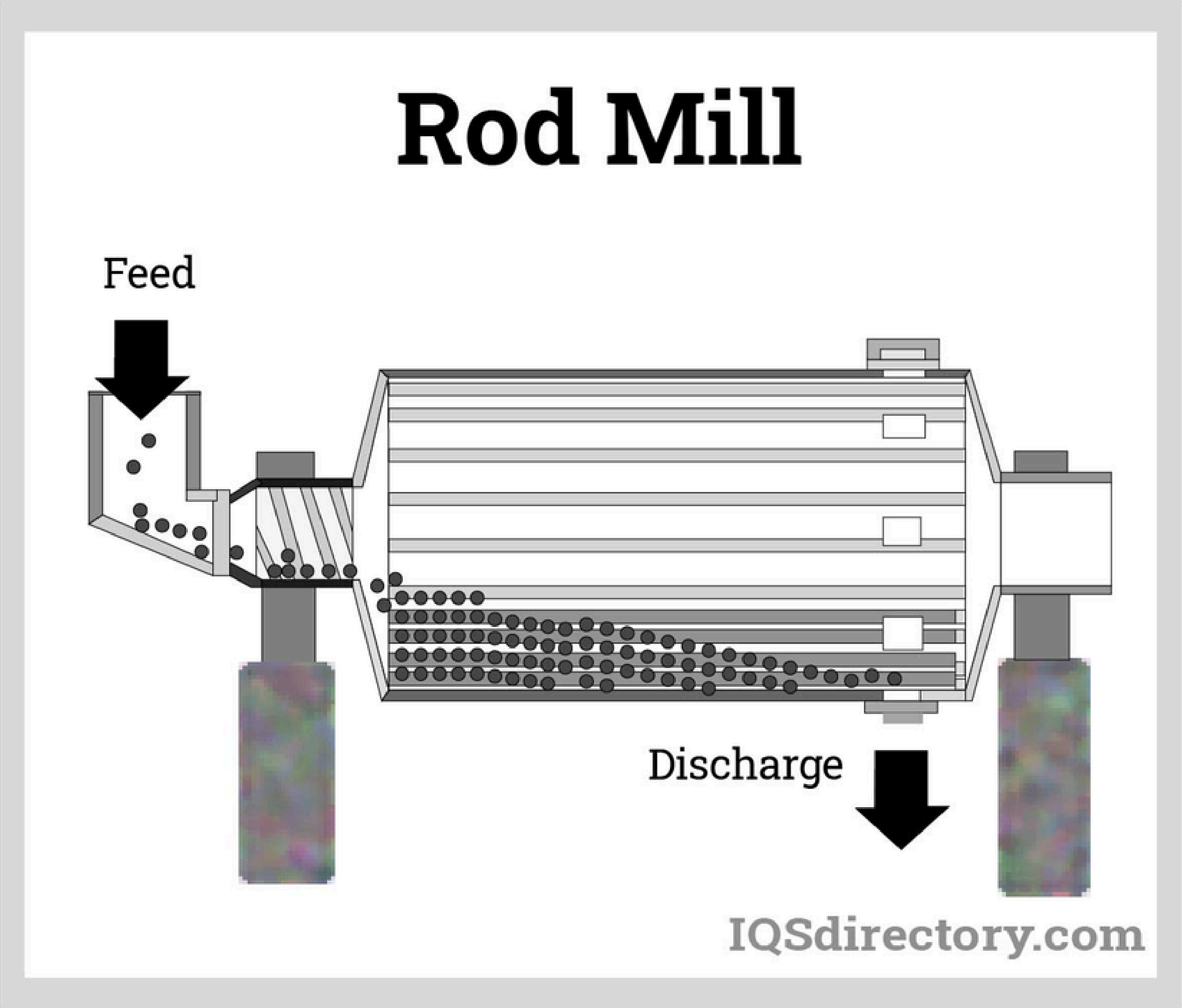
Attritor milling is more efficient than ball milling for obtaining small particle size and also results in less contamination.


Drive shaft
Water inlet
Input
Water outlet
Cooler
Rotating impeller
Heater
Medium inlet
Medium outlet

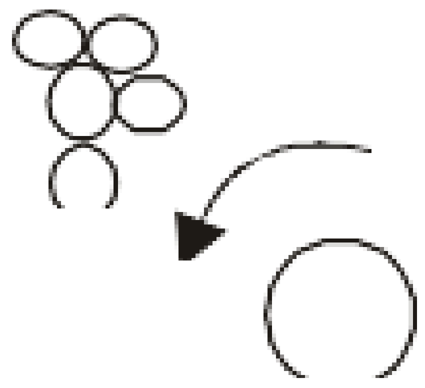
- **Rod mills** employ horizontal rods instead of balls to achieve grinding.

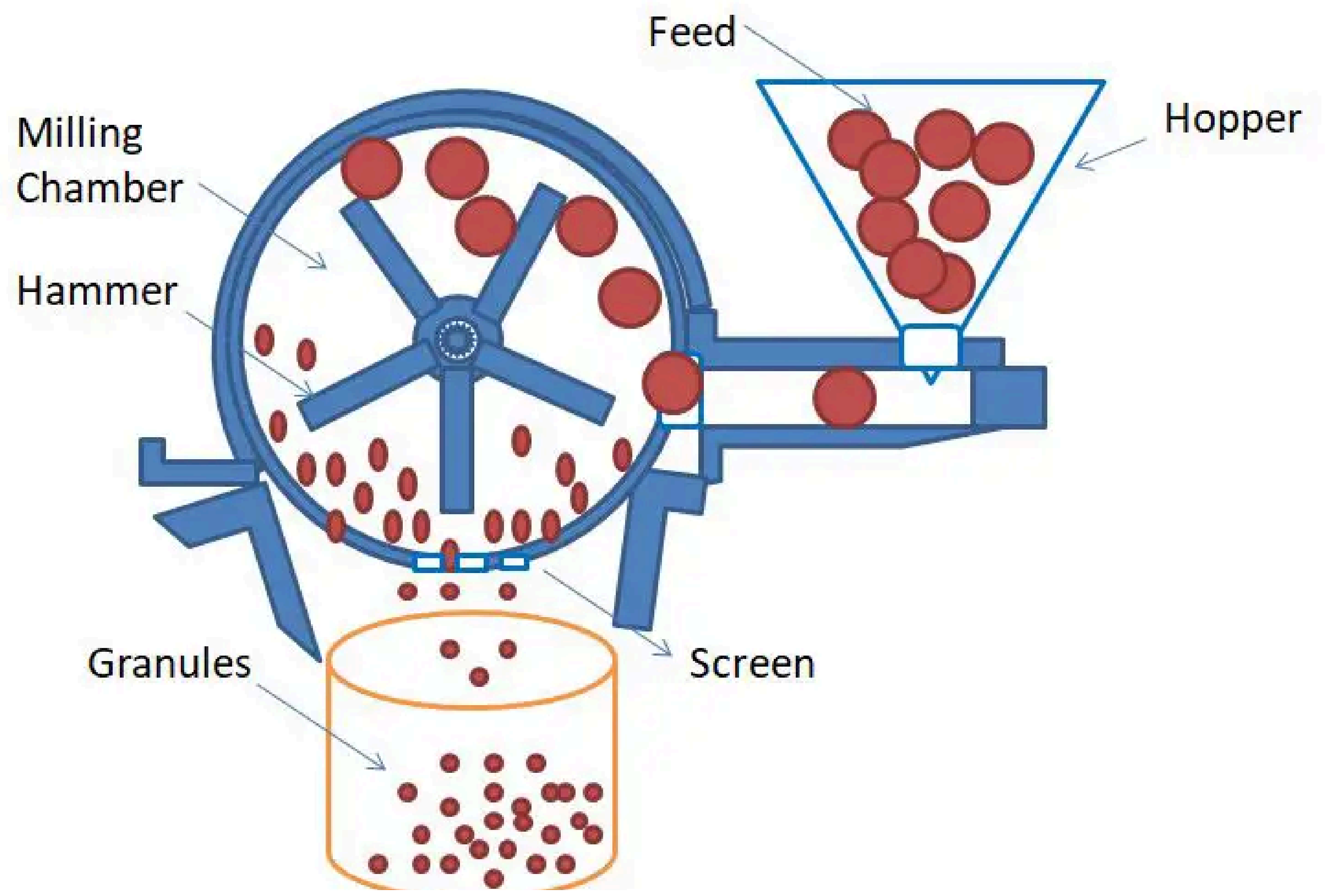
- The rod diameter, depending on the mill drum diameter and on the granularity of the discharge material, is in the range of 40 to 100 mm.

- The mill speed varies from 12 to 30 rpm.

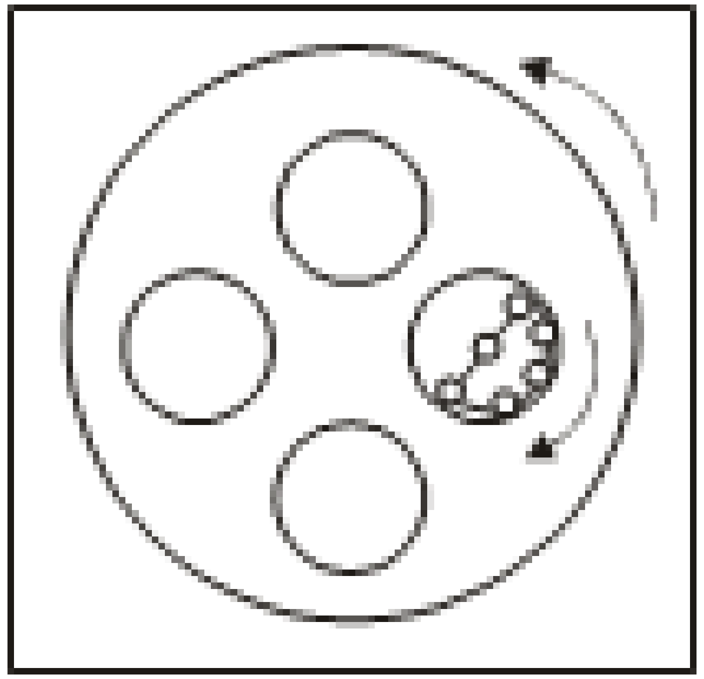


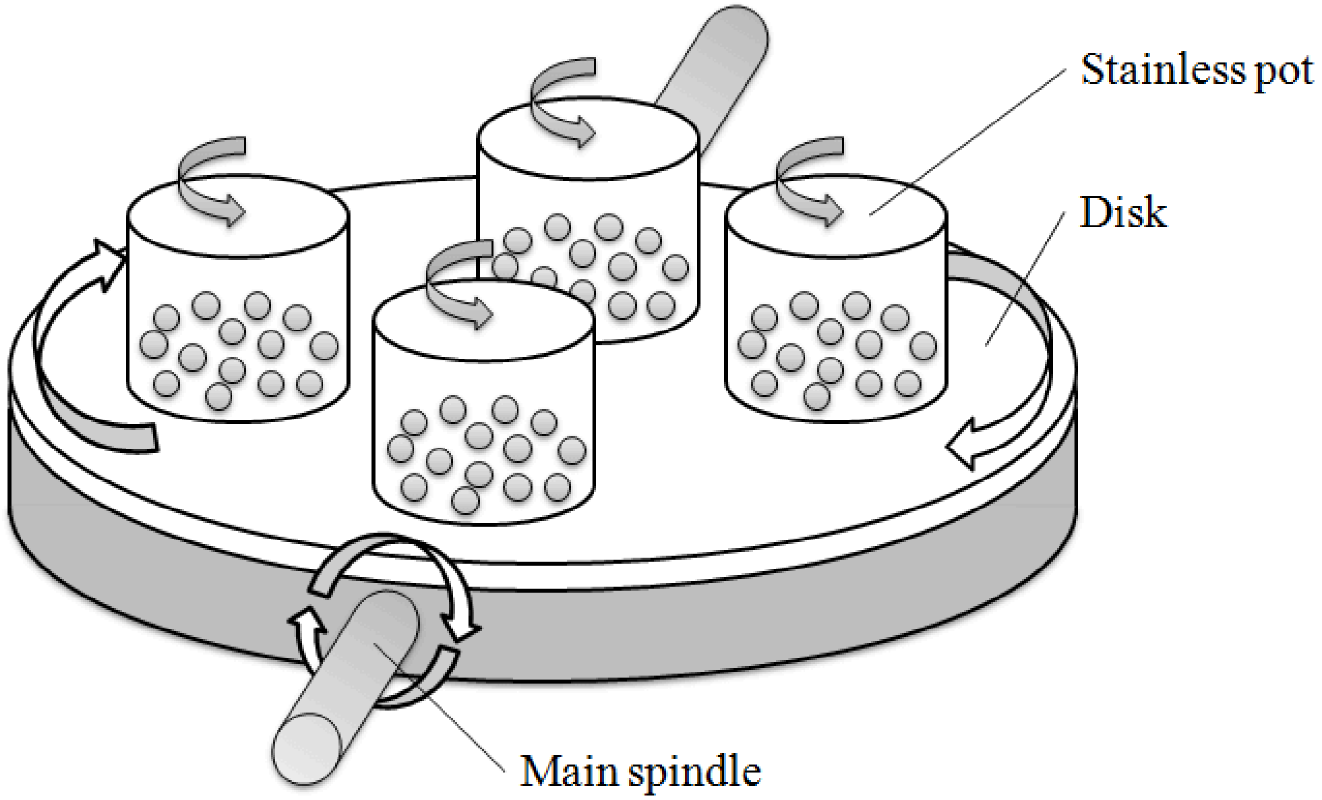
- **Hammer mills** employ blades to grind the material.



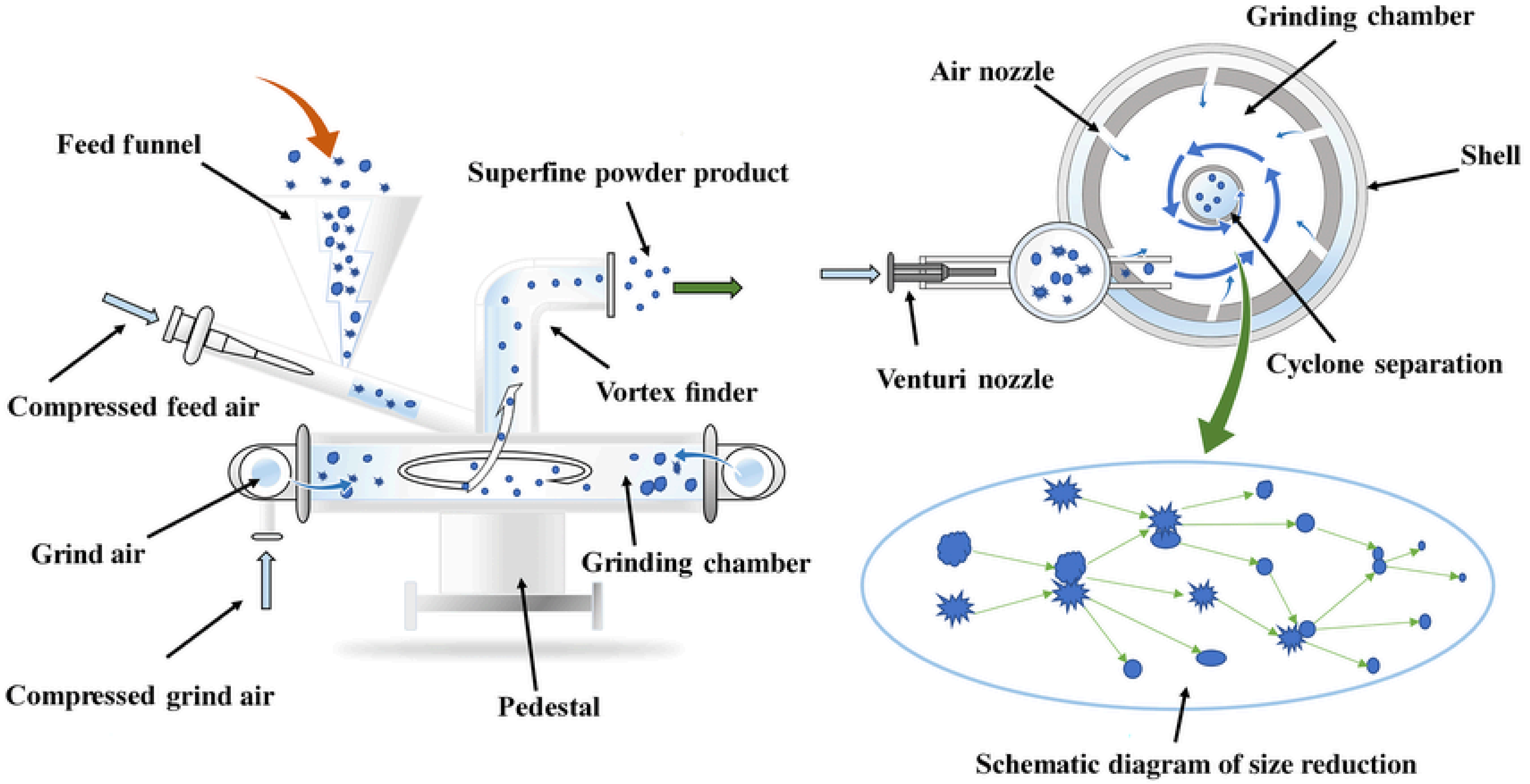


- **Planetary mill** is another high-energy mill widely used for production of metal, alloy and composite powders.



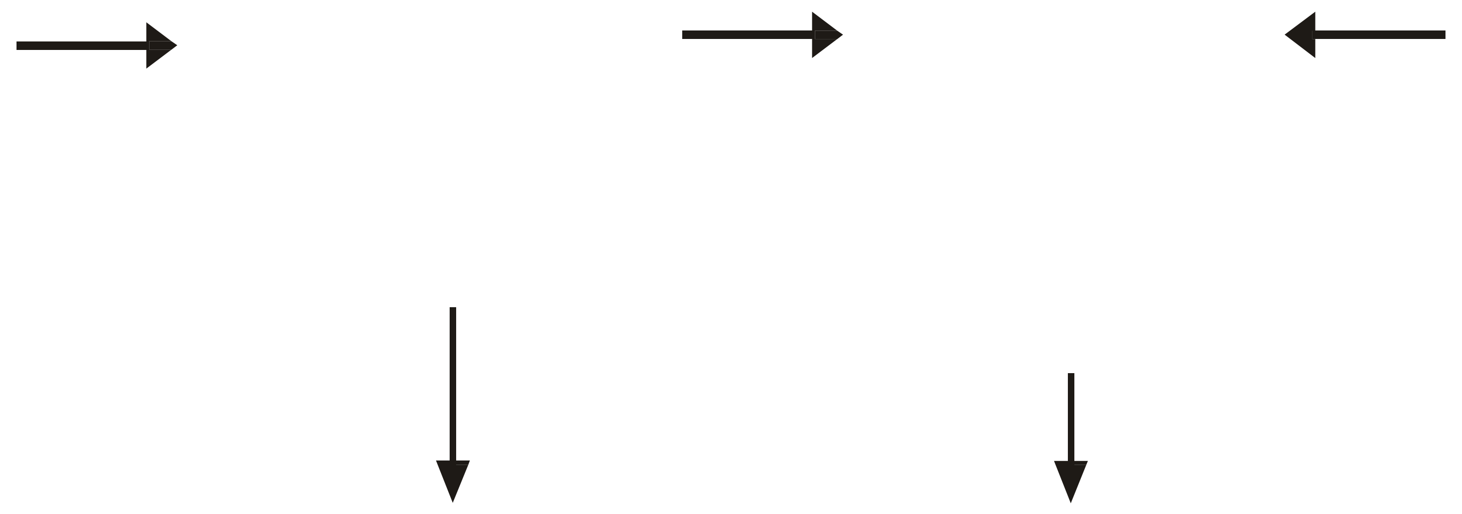


- **Vortex mills** find increasing applications in **fine and superfine grinding**.
- A vortex mill consists of a wear-resistant material-lined chamber with two propellers running at speeds up to 3,000 rpm.
- The rotation of the propellers creates a vortex, which causes the metal particles to get entrained from a bin.
- The propellers knock them against each other and grind them.
- The ground material is then withdrawn through the mill by a fan and allowed to settle in collection chambers.
- The granularity after grinding varies from 50 to 200 microns.
- When the particles collide, heating occurs, resulting in oxidation and agglomeration.
- In order to prevent heating up, the body of the mill is provided with a cooling jacket through which water is circulated during operation.
- To prevent oxidation of the powders, an inert gas is continuously passed into the crushing chamber.


Feed funnel
Superfine powder product
Compressed feed air
Vortex finder
Grind air
Grinding chamber
Compressed grind air
Pedestal
Grinding chamber
Air nozzle
Shell
Venturi nozzle
Cyclone separation
Schematic diagram of size reduction

Fluid Energy Grinding

- High-pressure fluid energy grinding or jet milling, is a versatile method of grinding solids continuously into very fine particles.



- The basic principle of fluid energy mill is to induce particles to **collide against each other at high velocity**, causing them to fracture into smaller particles.

- Multiple collisions enhance the reduction process, and therefore, multiple jet arrangements are normally incorporated in the mill design.

- The **fluid used is either air at about 0.7 MPa or steam at about 2 MPa.**

- In the case of **pyrophoric or volatile materials, either nitrogen or carbon dioxide is used as a protective atmosphere.**

- The pressurized fluid is introduced into the grinding zone through specially designed nozzles which convert the applied pressure into kinetic energy.

- **Feed material** (to be pulverized) is simultaneously introduced into this turbulent zone and accelerated with the expanding gas stream so that multiple particle- particle impact is induced at sonic or supersonic velocities.
- **Kinetic energy,** expressed as $mv^2/2$ forms the basis for all jet-milling design.
- The **velocity of the driving fluid exiting the nozzles is directly proportional to the square root of the absolute temperature** of the fluid entering the nozzle.

- Hence, to obtain maximum energy from a given mass of fluid, it is **preferable to raise the temperature of the incoming fluid** to the maximum possible level without affecting the feed material.

- It is probable that some of the particles may require **further comminution** after the initial impacting process.

- For this, the **oversize particles are separated** from the fines by a suitable classification process (e.g. **centrifugal classification**) for reprocessing.
- The fines are removed by the viscous drag of the exhaust gases to be carried away for collection, while the coarser particles are drawn away from the exit by centrifugal forces and recirculated into the grinding zone along with the fresh materials to be ground.
- Jet milling techniques are usually aimed at producing finest possible powders, with the finished product having a median particle size of less than **5 microns**.

- Hence, **jet mill feed** is generally prepared by **precrushing or mechanical milling to** a desired level (e.g. around 150 microns) to generate a finished product of around 44 microns.

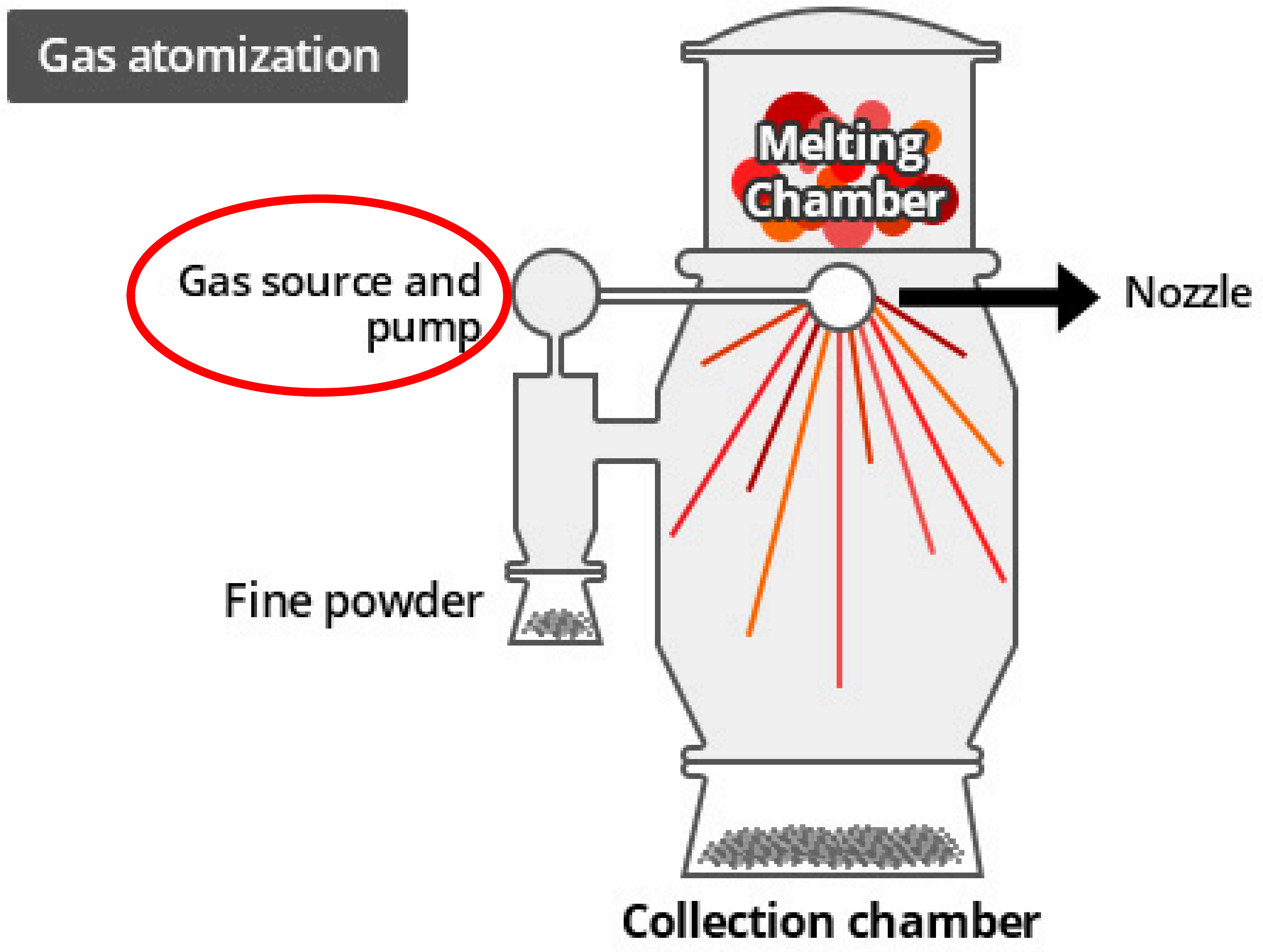
- **Uniform spheroidal or cubic particles tend to grind better** than needle-shaped or plate type particles due to the random orientation of the particles and consequent projected surface during the impact.

Machining

- **Magnesium, beryllium, silver, solder and dental alloy powders** are specifically made by machining.
- Turnings and chips obtained during machining are subsequently crushed or ground (to the desired degree of fineness) into powders.

Shotting

- In this method, a fine stream of **molten metal is poured through a vibratory screen into air or other protective gas medium**.
- When the molten metal falls through the screen, it disintegrates and solidifies as spherical particles.
- The particles generally get oxidized. The sizes of the particles obtained by this method depend on factors like temperature, gas used (like inert gas), aperture size of the screen and frequency of vibration of the screen.
- Typical examples of metals produced by this method include copper, brass, aluminium, zinc, tin, lead and nickel.
- Similar to gas atomisation but with little difference



Gas source and pump are absent in Shotting instead a vibratory sieve is introduced.

Sieve size controls the particle size.

It is coarser as compared to atomization

Graining

This process is exactly like shotting but the material falling through the **sieve is collected in water**.

Powders produced by this method include cadmium, bismuth and antimony.

Chemical Methods

- Chemical reactions to produce metal **powders directly from their compounds (e.g. oxides)** at temperatures well below the melting point of the metals concerned.
- Chemical reactions offer attractive means of producing metal powders having a wide range of characteristics like purity, size and shape.
- **The thermodynamic and kinetic aspects of the process are, however, very important in chemical methods.**
- These reactions are based on reduction of oxide, precipitation from solution or from a salt, chemical embrittlement and thermal decomposition of compounds.

Reduction

- Reduction of compounds, particularly oxides, by the use of reducing agents (solid or gaseous) is one of the most widely used and the oldest method of producing metal powders.
- Other compounds such as hydrides, oxalates and formates can also be used as starting material for reduction.
- This is a **convenient, economical and extremely flexible method to control the properties of the product such as size, shape and porosity over a wide range**.
- It is widely employed for producing powders of **iron, copper, nickel, cobalt, tungsten, molybdenum, tantalum, thorium and titanium.**
- Powders produced by this route exhibit **pores within each powder particle** and hence are called **sponge powders.**

- **This sponginess**, controlled by the amount and size of the pores, aids in **improved compaction and sintering characteristics.**

- Gases such as hydrogen, carbon monoxide, disassociated ammonia, coal gas, enriched furnace gas, natural gas, partially-combusted hydrocarbons, alkaline metal vapours, carbons as well as metals are used as reducing agents in gaseous reduction of compounds.

$$MO + H_2 \qquad M + H_2O$$
$$MO + CO \qquad M + CO_2$$

- Production of sponge iron powders.
- two different routes (i) the Hoeganaes process, and (ii) Pyron process.

- In the ***Hoeganaes process***, developed in Sweden, **pure magnetite** (Fe_3O_4) is crushed and **treated with coke and limestone in silicon carbide** containers at 1,200°C for times up to 24 hours.

$$Fe_3O_4 + CO \rightarrow 3FeO + CO_2\uparrow$$
$$FeO + CO \rightarrow Fe + CO_2\uparrow$$
$$FeO + C \rightarrow Fe + CO\uparrow$$

- **Limestone acts as a flux** and prevents sulphur from contaminating the iron powders.
- The resulting spongy mass is then crushed to fine sizes and reduced in a hydrogen atmosphere in a mesh belt furnace..

- In the ***Pyron process***, mill scale is used as the starting material.
- This process involves the conversion of **magnetite to hematite** to facilitate subsequent **reduction by hydrogen**.
- The reduction is carried out at a temperature of 980°C.

$$3Fe_2O_3 + CO \rightarrow 2Fe_3O_4 + CO_2 \uparrow$$
$$Fe_3O_4 + H_2 \longrightarrow 3FeO + H_2O \uparrow$$
$$FeO + H_2 \longrightarrow Fe + H_2O \uparrow$$

- In cases where **oxides are not reducible by carbon or hydrogen**, the reduction is carried out using alkali or alkaline earth metal. For example, titanium oxide can be reduced by a **calciothermic reduction.**

$$TiO_2 + 2Ca \quad\quad Ti + 2CaO$$

- reaction is **highly exothermic** and difficult to control, **calcium hydride** is used as a reducing agent.

$$TiO_2 + CaH_2 \quad\quad Ti + CaO + H_2O$$

- Calcium hydride acts as an extremely powerful reducing agent as it dissociates at temperatures between 800–900°C into nascent hydrogen and metallic calcium.
- Reduction is achieved by mixing calcium and titanium hydride powder in proper proportion and heating in hydrogen or vacuum at high temperatures (higher than the melting point of the Cu-Ti eutectic alloy).
- The resulting molten alloy is cooled to room temperature and crushed to obtain the powder.

The generalized equation for reduction by calcium hydride may be written as:

$$MXOy + yCaH_2 \quad XM + yCa + yH_2O$$

- Many metals have been prepared by metallic reduction of oxides.
  Chromium powder is prepared by the reduction of chromium oxide with magnesium

  Zirconium powder is prepared by the reduction of zirconium oxide with magnesium (or calcium) in an inert gas atmosphere.

- **In all the reduction methods, control of powder characteristics** like particle size, shape, apparent density and the purity of the final product is achieved by the control of **starting material as well as process parameters.**
- Raw material characteristics include purity, particle size and shape.
- Important process parameters are temperature and time of reduction as well as the type of reducing agent.
- For instance, lower reduction temperature results in fine powder but takes longer times to produce the powders.
- On the other hand, higher reduction temperature may result in faster reduction, but powders may sinter and form agglomerates.
- Other important parameters include the pressure and flow rate of the gas in the case of gaseous reduction.
- When hydrogen is used as reducing agent, moisture must be removed to avoid undesirable reactions.

Thermal Decomposition

- **Metal carbonyls are formed by passing carbon monoxide at high pressure** over heated metal sponge and condensing the vapour as liquid.
- These are volatile liquids and usually stored as liquid under pressure.
- Decomposition of carbonyls of cobalt, iron, nickel, molybdenum and tungsten has been used to produce fine powders with high purity.
- The liquid is purified by fractional distillation and is then decomposed in the temperature range of 150 to 400°C at atmospheric pressure to produce fine, high-purity metal.
- The carbon monoxide released during decomposition is recycled.

- In the case of iron, iron powders are first treated with carbon monoxide to form iron pentacarbonyl according to the reaction
- $$Fe + 5\,CO \qquad Fe(CO)_5$$
- This carbonyl is subsequently decomposed at 200–220°C at atmospheric pressure to yield high purity (99.5%) spherical powders of iron according to the reaction.
- $$Fe(CO)_5 \qquad Fe + 5\,CO$$
- The carbon dioxide released in the process is recovered and sent for recirculation.
- To ensure purity and suitable particle shape, **reaction conditions are optimized so that the decomposition is spontaneous within the hot zone** and nucleation of powders does not occur on the cooler regions of the furnace (e.g. walls of the container).

- Too low a decomposition temperature would result in slow and incomplete decomposition
- Too high a temperature (> 400°C) would lead to oxidation.
- **Iron powders produced by this technique are perfectly spherical** and have high purity (99.5%) containing only trace amounts of carbon, oxygen and nitrogen.
- Iron powder produced by this method is a good soft magnetic material and is used to make magnetic cores.
- **Nickel powders produced by this technique are usually irregular in shape**, porous and fine.
- These powders are used in the production of alkaline battery and fuel cell electrodes.
- **Iron-nickel alloy powder produced by thermal decomposition of a mixture containing iron and nickel carbonyl vapours has been used in industry to manufacture high- permeability cores for communication applications**.

Other typical carbonyl reactions

Nickel:

$Ni + 4CO \quad Ni(CO)_4$ —— formation of carbonyl
$Ni(CO)_4 \quad Ni + 4CO$ —— decomposition of carbonyl

Tungsten:

$W + 6CO \quad W(CO)_6$ —— formation of carbonyl
$W(CO)_6 \quad W + 6CO$ —— decomposition of carbonyl

Cobalt:

$2Co + 8CO \quad Co_2(CO)_8$ —— formation of carbonyl
$Co_2(CO)_8 \quad 2Co + 8CO$ —— decomposition of carbonyl

Hydride–De hydride Process

- Decomposition of metal hydride is also an effective method for producing high-purity powders.
- This process is also known as hydride-de hydride process and is mainly used for the production of reactive metals like titanium, vanadium, niobium, hafnium, zirconium, thorium and uranium.

  - The process is carried out in two stages:
  - In the first stage, the reactive metal is converted to its hydride (mono or dihydride) by reaction with hydrogen at a suitable temperature under pressure.
  - In the case of titanium, the reaction with hydrogen takes place in the temperature range of 300°C–500°C and can be written as:
- $$Ti + H_2 \quad TiH_2$$

The resulting hydride is very brittle and can be easily crushed to powder of required fineness.

In the second stage, the crushed hydride is decomposed to metallic powder by heating under vacuum at the same temperature at which the hydride was formed.

$$TiH_2 \quad Ti + H_2$$

- The hydride-de hydride process is also used for very fine uranium powder production. The process involves heating uranium alloy ingot at a temperature of 300°C in a hydrogen atmosphere to form uranium hydride. After hydriding is complete, the uranium hydride is reduced by heating the powder under vacuum. The hydrogen dissociates from the powder leaving high-purity uranium alloy in powdered form. The reactions are:
    - $3H_2 + 2U \quad 2UH_3$ —— hydriding reaction
    - $2UH_3 \quad 3H_2 + 2U$ —— dehydriding reaction

Precipitation from Aqueous Solution

- Metal precipitation from solution can be carried out by electrolysis, sedimentation or chemical reduction.
- Precipitation may be achieved by precipitating a compound of the metal such as hydroxide, carbonate or oxalate followed by heating which results in decomposition and reduction.

- **A metal may also be precipitated from its aqueous solution by the addition of a less noble metal.**
- This produces very fine metal powders having low-apparent density.
  - Iron, copper and nickel powders are obtained by adding aluminium to their sulphate solution containing $HgCl_2$.
  - An activating agent is usually used to aid the precipitation process.
  - Gold powder is precipitated with zinc and copper while copper powder is precipitated from its sulphate solution by the addition of iron.

Hydrometallurgical or Gaseous Reduction Method

- Precipitation of metals from salt solution can also be achieved using gaseous reductants like hydrogen.
- Metal powders particularly **copper, nickel and cobalt** are precipitated on commercial scale by hydrometallurgical method by the reduction of aqueous solution or slurries of salts of metal with hydrogen.
- The basic principle is that a metallic ions like nickel, copper or cobalt react with hydrogen according to the following reaction:

$$M^{++} + H_2 \qquad M + 2H^{+}$$

  - If the solution is ammoniacal, the equation becomes
- $$M^{++} + 2NH_3 + H_2 \qquad M + 2NH_4$$

- An important example of this technique is the **Sheritt-Gordon process** used for nickel powder production.

- In this method, an **ammoniacal nickel sulphate solution** - by leaching is treated with hydrogen under pressure (1.4 MPa) at a temperature of 200°C.
- A **catalyst, usually ferrous sulphate** is employed for precipitation of nickel.
- After precipitation, the lean solution is poured off and fresh solution is added, and the process is repeated several times.
- The resulting powders are then washed and dried.
- Coprecipitation of different metals from solution allows the production of composite powders.
- **Precipitation from the vapour phase is also used to produce metal powder.**
- **For example, fine spherical powders of tungsten and molybdenum can be obtained from their gaseous hexafluorides.**

Precipitation from Fused Salt

- Powders particularly of **reactive metals** can be prepared by precipitating from fused salt.
- For producing **zirconium** powder, zirconium tetrachloride salt is mixed with equal amount of potassium chloride and some magnesium.
- Magnesium replaces zirconium when heated to 750°C and powder particles of zirconium settle at the base of the treatment chamber.
- In another method, zirconium tetrachloride vapour is introduced into a salt bath of sodium chloride and potassium chloride to produce zirconium.
- Other metals produced by this method include **beryllium and thorium.**
- Several **intermetallic compounds** have also been prepared in powder form by reacting two amalgams.

Reduction–Condensation Method

- Very fine **zinc oxide** is mixed with powdered charcoal and heated until zinc vapour is formed by the reaction of the oxide with carbon monoxide.
- Zinc vapour is condensed in the cooler regions of the retort.
- However, because of its high-oxygen content the powders are not considered suitable for powder metallurgy purposes.
- It is possible to obtain very fine powder of high purity through the use of a second condenser.
- **Magnesium powder** is produced by this method by reducing magnesia with calcium carbide followed by collection of the magnesium vapour in oil.
- This method is also used to prepare **cadmium powder of size less then 500Å**.

Physical Methods

Electrolytic Deposition

- Metal powders can be produced by electro deposition from aqueous solution and fused salts.
- In this method, the processing conditions are so chosen that **metals of high purity are precipitated from aqueous solutions on the cathode** of an electrolytic cell.
- This method is mainly used in the production of **copper and iron**, although metal powders of zinc, tin, nickel, cadmium, antimony, silver, lead and beryllium can also be prepared.
- Fused salt electrolysis has also been used for the production of tantalum and beryllium powders.

- Electrodeposition of iron and copper powder generally **use sulphate solutions.**
- However **chlorides, cyanides of** these metals also have been used successfully.
- Addition of ammonium chloride to iron sulphate or chloride increases the conductivity and reduces oxidation of the powders.
- **Oxygen content of electrolytic powder is generally high**.
- The powder deposits are subsequently pulverized by milling in hammer mill.
- **The milled powders are then annealed in hydrogen atmosphere to remove oxygen.**
- The electrolytic powders are crystalline and are **generally dendritic (fern like shape)** with low apparent density and **flow rate favourable for pressing.**
- **Preparation of alloy powder is also possible by the use of a mixture of electrolytes with two or more metal salts.**
- **Electrolytic bronze and brass powders have been produced by such a method.**

- **Powders of thorium, tantalum and vanadium** have been prepared by fused salt electrolysis.
- In electrodeposition, **depending upon the processing conditions, the final product may be obtained in three different forms**.

  A hard brittle layer of pure metal which is subsequently milled to obtain powder (e.g. iron powder).
  A soft, spongy substance which is loosely adherent and easily removed by scrubbing.
  A direct powder deposit from the electrolyte which collects at the bottom of the cell.

- **The last two methods are extensively used for the production of large quantities of powder.**

The factors that promote powdery deposits

- High-current density.
- Low-metal concentration.
- pH of the bath.
- Low temperature.
- High viscosity (use of surfactants).
- Circulation of electrolyte to avoid or suppress convection.

By a careful control of these factors, a close control over the characteristics of the final powder is possible.

- The **selection of a suitable cathode** material permitting the complete removal of the deposits is of great importance for continuous commercial production.
- **Highly polished and smooth surfaces are desirable** so that the deposit is stripped or removed by flexing (bending) or scrubbing or by rotating the cathode or by rapid circulation of the bath.
- **Electrolytic powders are usually given an annealing treatment** to reduce their reactivity and their brittleness (due to pick up of gases).
- Annealing is carried out under **reducing conditions** (hydrogen or hydrocarbon gas atmosphere).

- **Advantages:**
- Powders are of high purity with excellent compactibility and sinterability.
- Allows the production of wide range of powder quality by altering the bath composition.

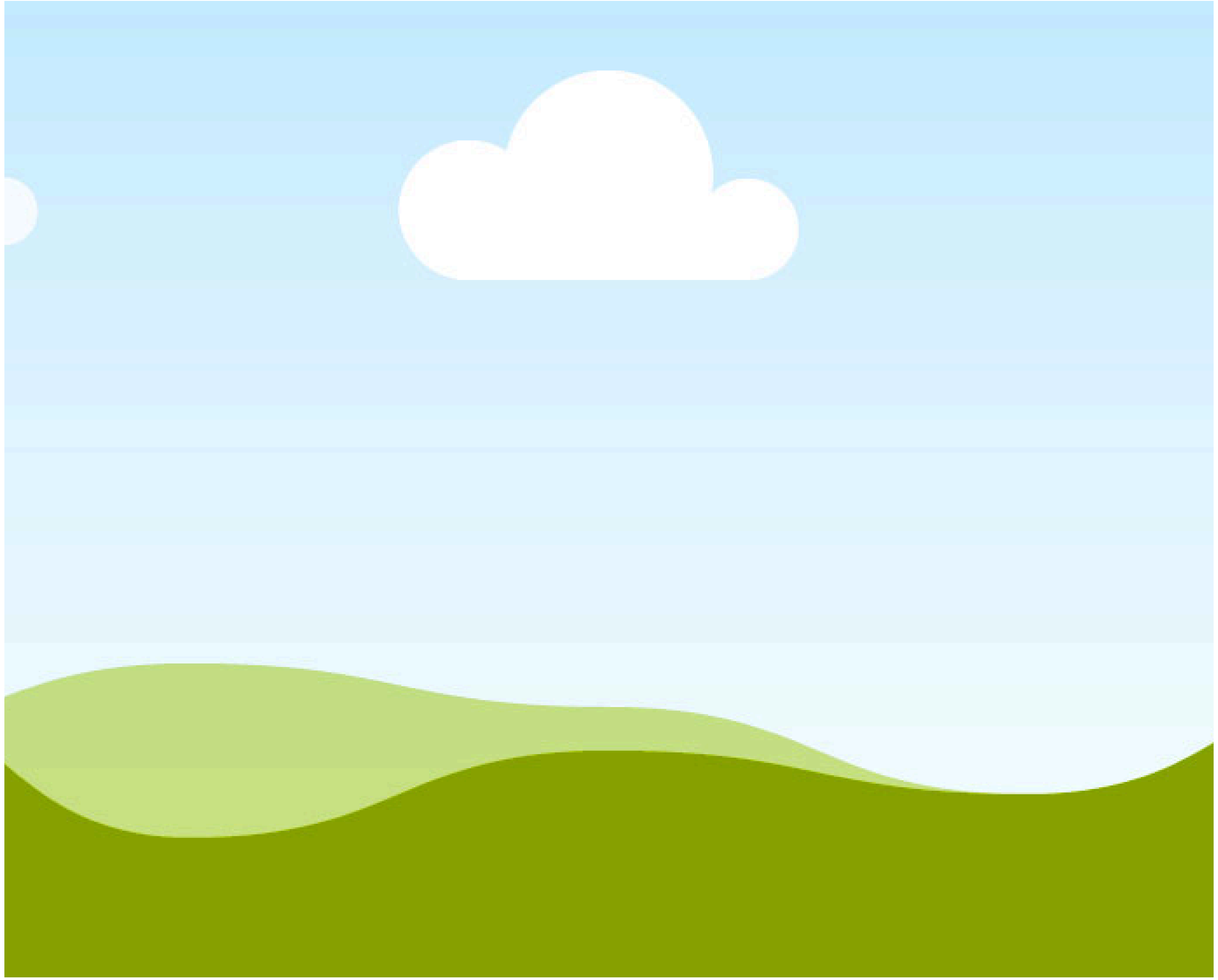
- **Disadvantages**
- The process is time-consuming.
- Deposited powder must be washed thoroughly, dried and annealed to avoid rapid oxidation, which adds to the operating cost.
- Handling and processing involves toxic chemicals resulting in pollution of work place.
- Waste disposal is another problem.

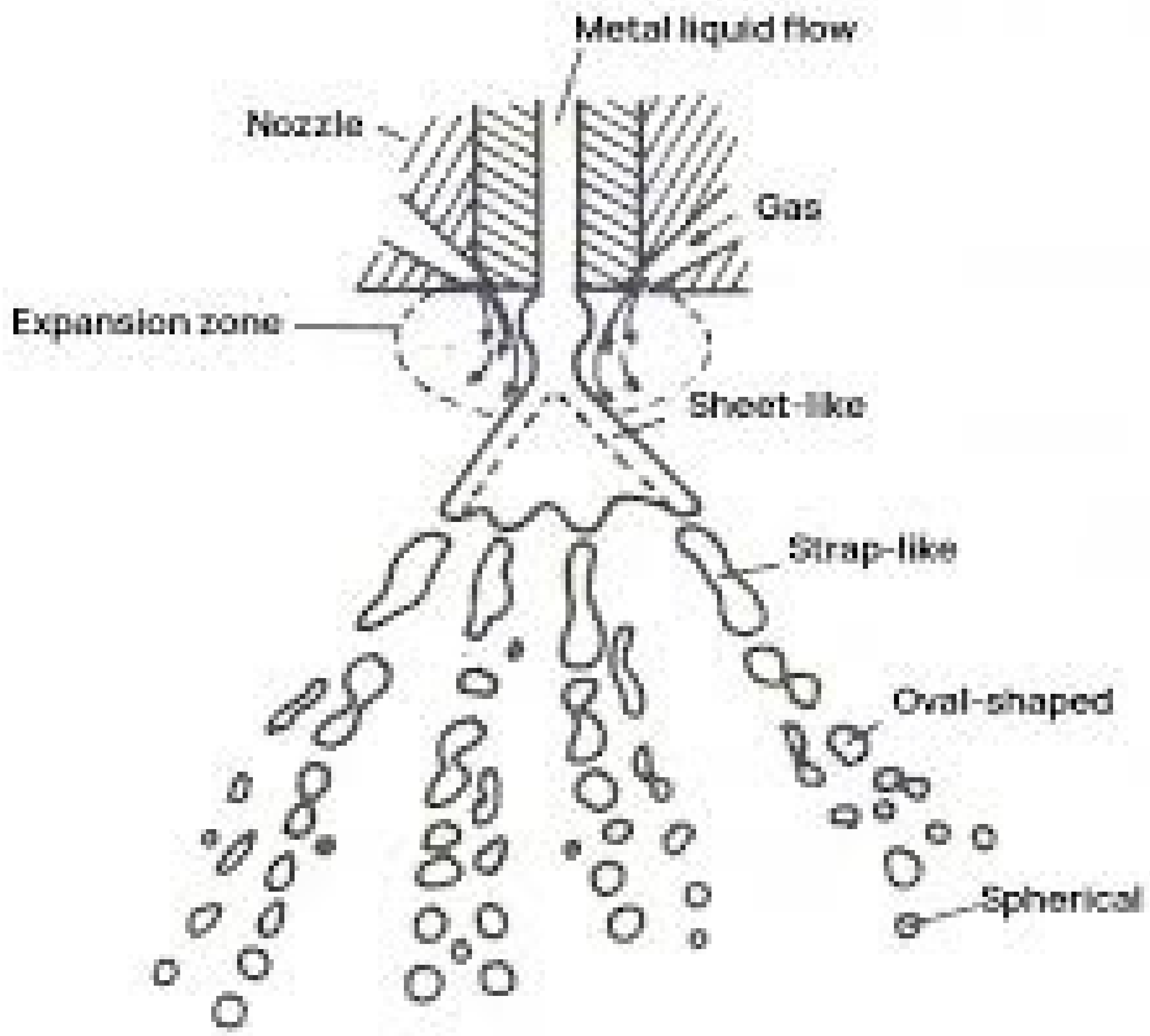
Atomization

- Atomization is one of the most widely used method of making powders of metals and alloys.
- This method accounts for major volume of powders produced for P/M applications.
- The major steps involved in atomization are melting of the metal/alloy, atomizing (disintegrating) the molten stream and solidifying the droplets.
  - Atomization as a powder production technique enables the production of **powders with crystalline and amorphous structures, alloys of immiscible systems as well as powders with specific characteristics**.

Mechanism of atomization

- In conventional (gas or water) atomization, a liquid metal is produced by pouring molten metal through a tundish with a nozzle at its base.
- The stream of liquid is then broken into droplets by the impingement of **high pressure gas or water.**
- **This disintegration of liquid stream has five stages**
- i) Formation of wavy surface of the liquid due to small disturbances
- ii) Wave fragmentation and ligament formation
- iii) Disintegration of ligament into fine droplets
- iv) Further breakdown of fragments into fine particles
- v) Collision and coalescence of particles





Metal liquid flow
Nozzle
Gas
Expansion zone
Sheet-like
Strap-like
Oval-shaped
Spherical

- The interaction between jets and liquid metal stream begins with the creation of small disturbances at liquid surfaces, which grow into shearing forces that fragment the liquid into ligaments.
- The broken ligaments are further made to fine particles because of high energy in impacting jet.
- Lower surface tension of molten metal, high cooling rate => formation of irregular surface => like in water atomization
- High surface tension, low cooling rates => spherical shape formation => like in inert gas atomization

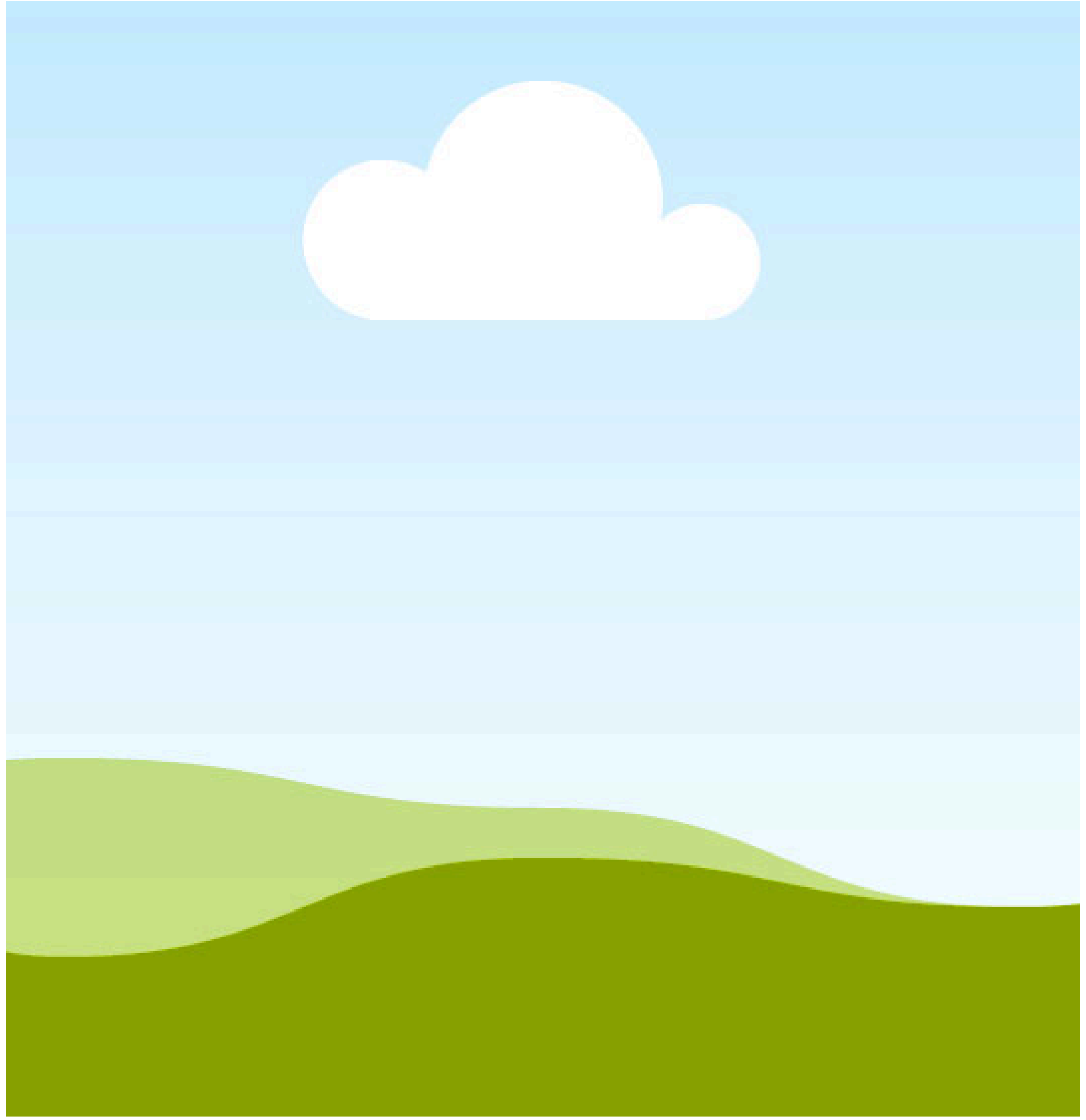
- The liquid metal stream velocity, **$v = A\ [2g\ (P_i - P_g)\rho]0.5$**
- where Pi – injection pressure of the liquid, Pg – pressure of atomizing medium, ρ – density of the liquid

Types

Water atomization
Gas atomization
Soluble gas or vacuum atomization
Centrifugal atomization
Rotating disc atomization
Ultrarapid solidification process
Ultrasonic atomization

Water atomization

- High pressure water jets are used to bring about the disintegration of molten metal stream.
- Water jets are used mainly because of their higher viscosity and quenching ability.
- This is an inexpensive process and can be used for small or large scale production.
- **But water should not chemically react with metals or alloys used.**



- Fit for high volume and low cost production
- Powders of average size from 150 micron to 400 micron
- Cooling rates from 10^3 to 10^5 K/s.
- Rapid extraction of heat results in irregular particle shape => less time to spheroidize in comparison to gas atomization
- Water pressure of 70 MPa for fine powders in 10 micron range

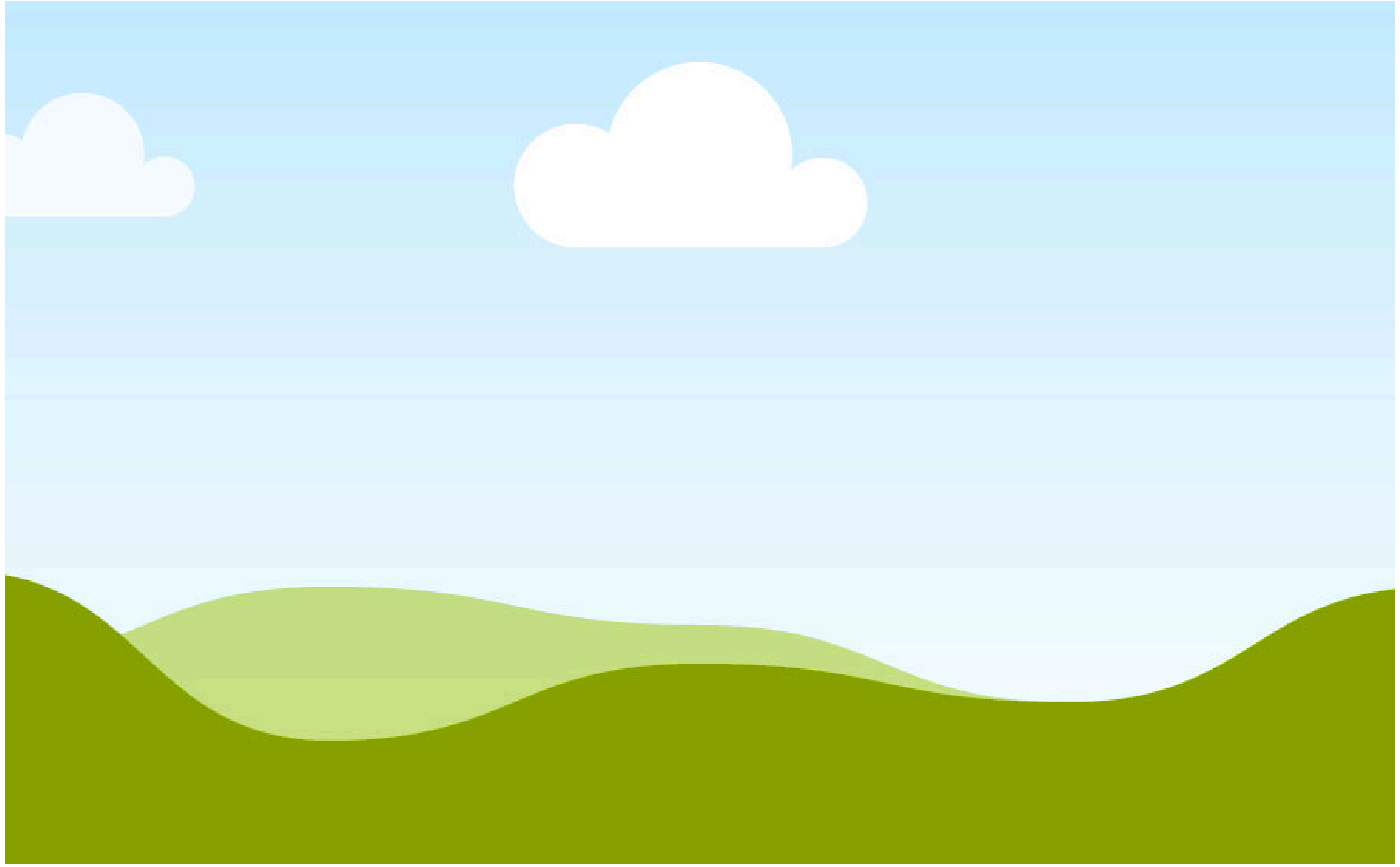
Important parameters:

1. Water pressure: Increase water pressure => size decrease => increased impact
2. water jet thickness: increase thickness => finer particles => volume of atomizing medium increases
3. Angle of water impingement with molten metal **(Apex angle)** & distance of jet travel

(Inert) Gas atomization

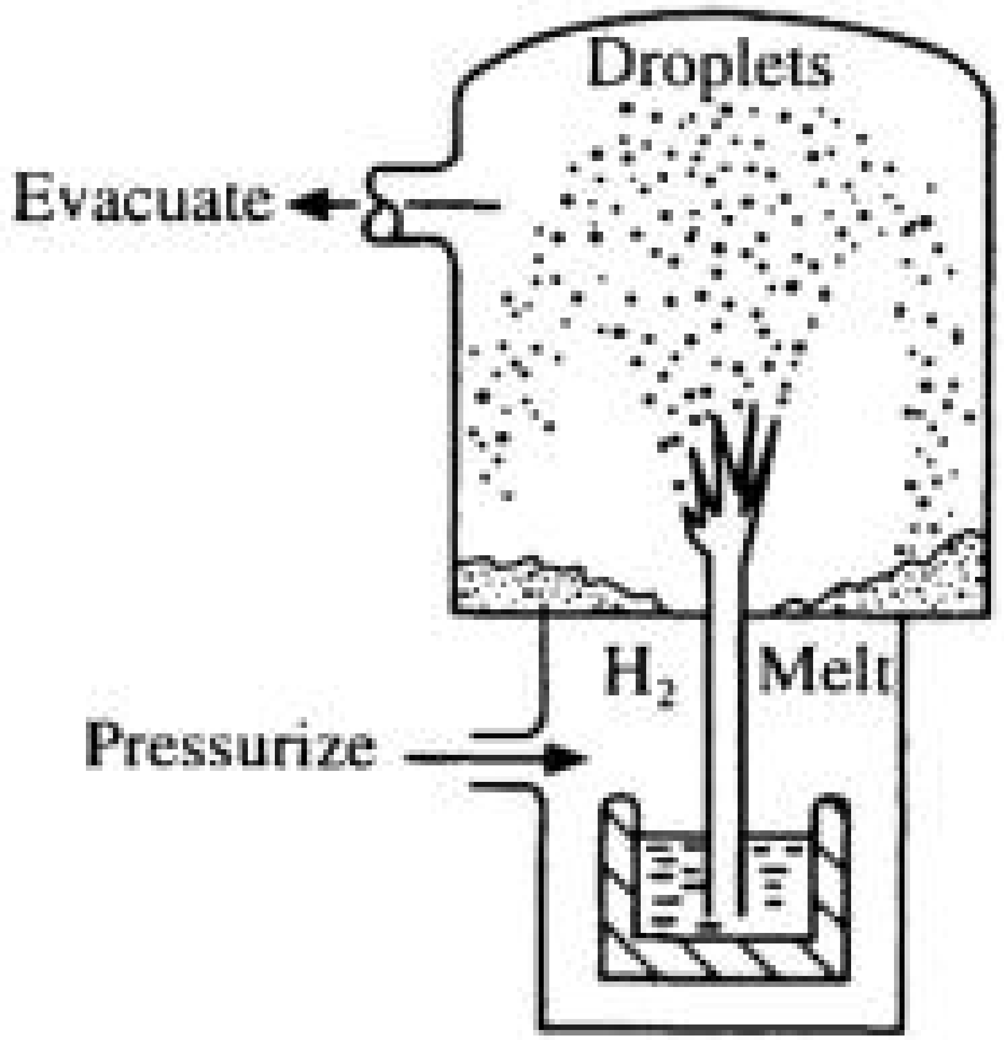
- Here instead of water, high velocity argon, nitrogen and helium gas jets are used.
- The molten metal is disintegrated and collected as atomized powder in a water bath.
- Fluidized bed cooling is used when certain powder characteristics are required.
- Production of high grade metal powders with spherical shape, high bulk density, flowability along with low oxygen content and high purity
- - Eg. Ni based super alloys

Horizontal gas atomization



Vacuum atomization

- when a molten metal supersaturated with a gas under pressure is suddenly exposed into vacuum, the gas coming from metal solution expands, causing atomization of the metal stream.
- This process gives very high purity powder.
- Usually hydrogen is used as gas.
- Hydrogen and argon mixture can also be used.

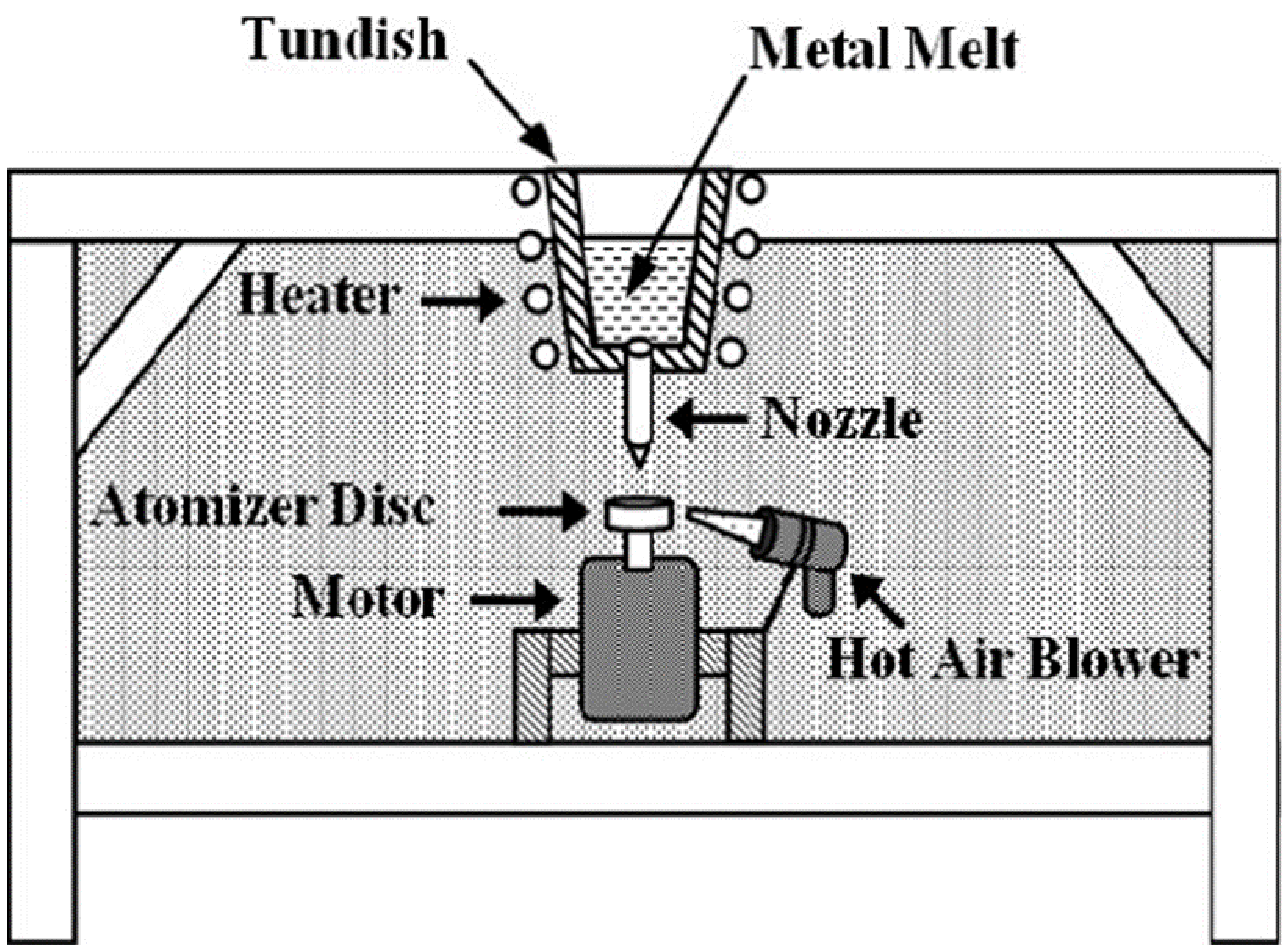


Centrifugal atomization

- One end of the metal bar is heated and melted by bringing it into contact with a non-consumable tungsten electrode, while rotating it longitudinally at very high speeds.
- The centrifugal force created causes the metal drops to be thrown off outwards.
- This will then be solidified as spherical shaped particles inside an evacuated chamber.
- Titanium powder can be made using this technique

Rotating disc atomization

- Impinging of a stream of molten metal on to the surface of rapidly spinning disc.
- This causes mechanical atomization of metal stream and causes the droplets to be thrown off the edges of the disk.
- The particles are spherical in shape and their size decreases with increasing disc speed.



Ultrarapid solidification processes

- A solidification rate of $100^{0}C/s$ is achieved in this process.
- This results in enhanced chemical homogeneity, formation of metastable crystalline phases, amorphous materials.

Atomization Unit

Melting and superheating facility:

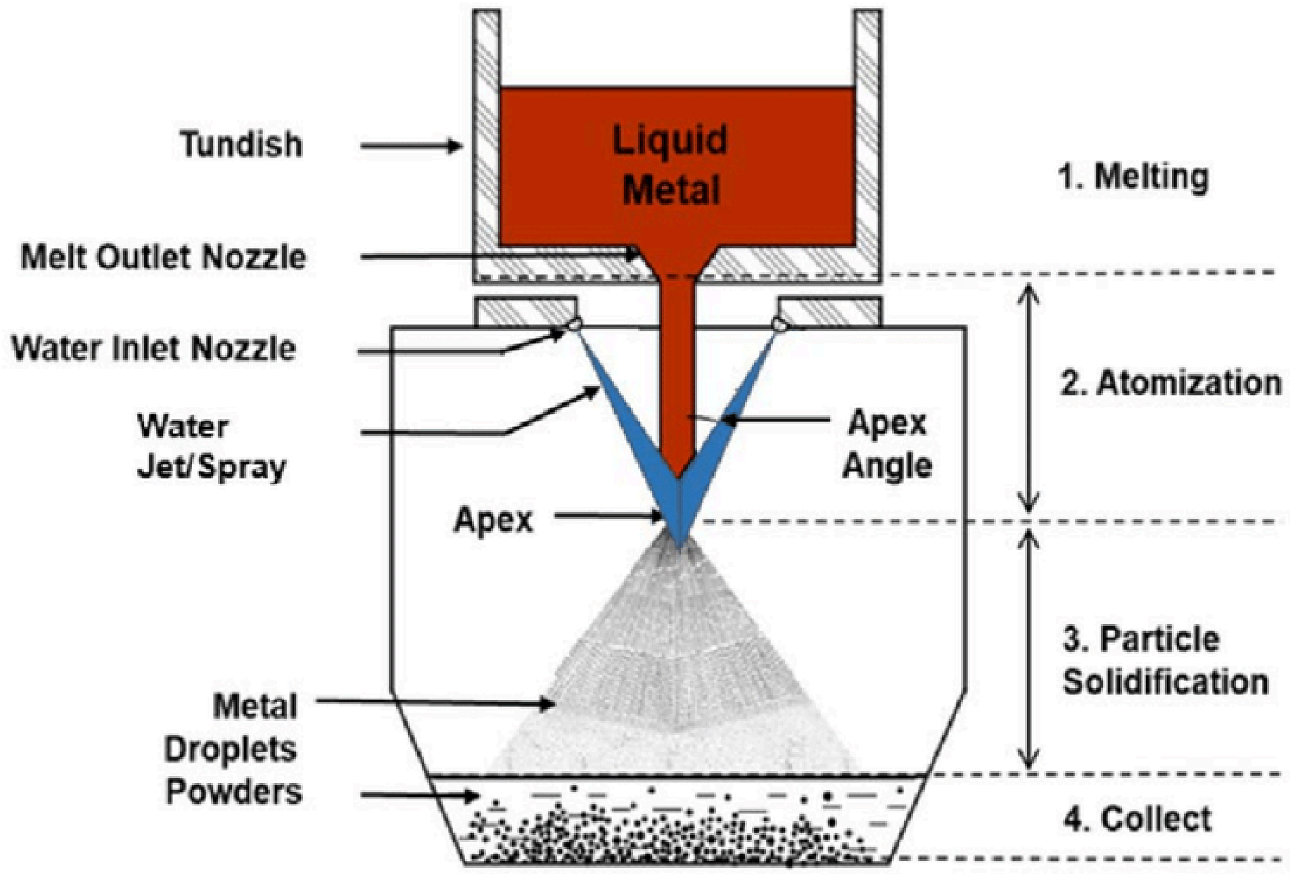
Standard melting furnaces can be used for producing the liquid metal.
This is usually done by air melting, inert gas or vacuum induction melting.
Complex alloys that are susceptible to contamination are melted in vacuum induction furnaces.
The metal is transferred to a tundish, which serves as reservoir for molten metal.

Atomization chamber:
An atomization nozzle system is necessary.
The nozzle that controls the size and shape of the metal stream if fixed at the bottom of the atomizing chamber.
In order to avoid oxidation of powders, the tank is purged with inert gas like nitrogen.

Powder collection tank:
The powders are collected in tank.
It could be dry collection or wet collection.
In dry collection, the powder particles solidify before reaching the bottom of the tank.
In wet collection, powder particles collected in the bottom of the water tank.
The tank has to be cooled extremely if used for large scale production.
During operation, the atomization unit is kept evacuated to 10^{-3} mm of Hg, tested for leak and filled with argon gas

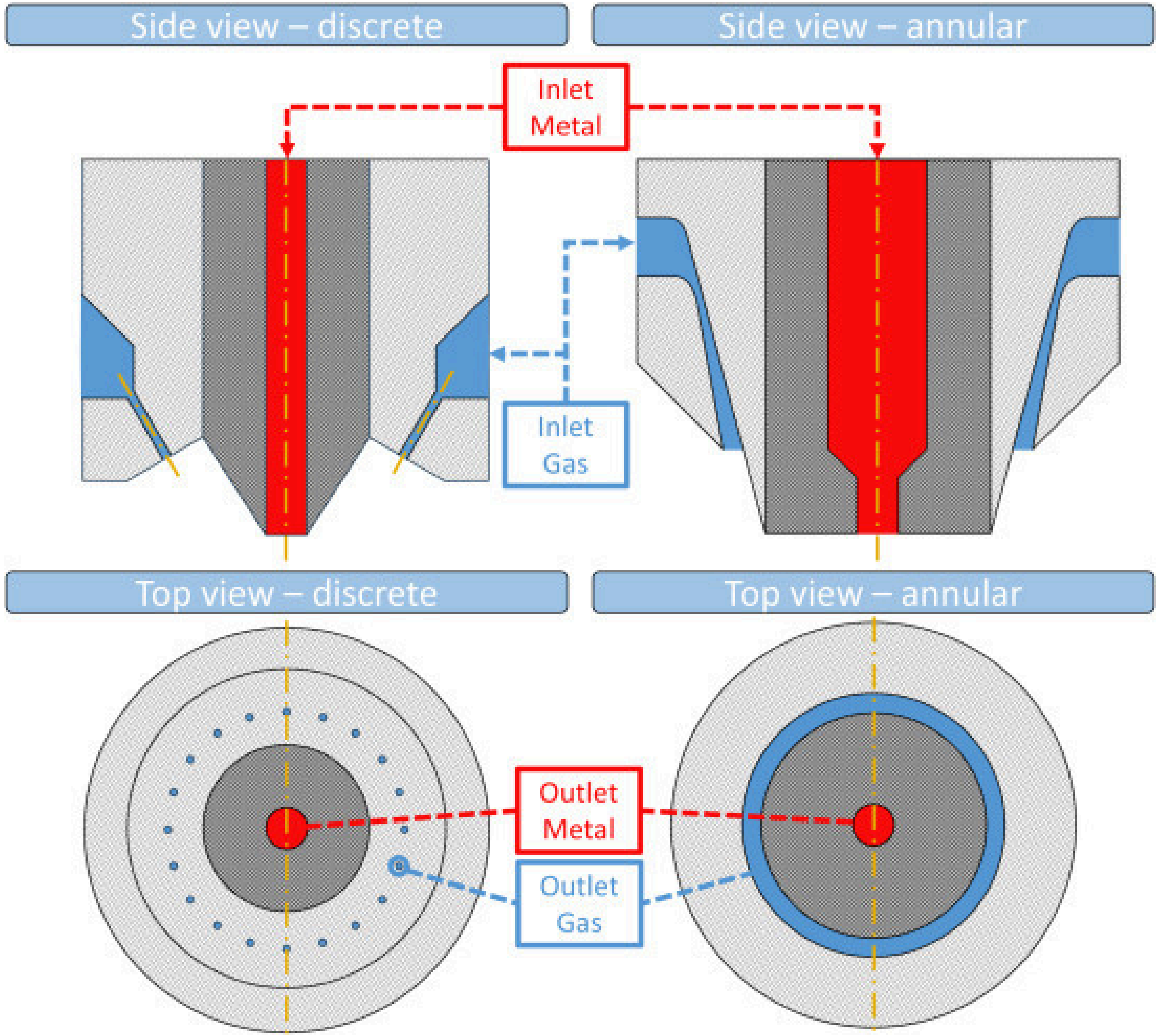
Atomizing nozzles

- Function is to control the flow and the pattern of atomizing medium to provide for efficient disintegration of powders
- For a given nozzle design, the average particle size is controlled by the pressure of the atomizing medium and also by the apex angle between the axes of the gas jets
- Higher apex angle lead to smaller particle size
- Apex angle for water atomization is smaller than for gas atomization



Nozzle design:

- i) free falling, ii) confined design
- **In free falling**, molten metal comes in contact with atomizing medium after some distance.
- This is mainly used in water atomization.
- **In confined design** used with annular nozzle, atomization occurs at the exit of the nozzle.
- Gas atomization is used generally for this.
- This has higher efficiency than free falling type.
- **One has to be cautious that "freeze up" of metal in the nozzle has to be avoided.**



Side view – discrete
Side view – annular
Inlet Metal
Inlet Gas
Top view – discrete
Top view – annular
Outlet Metal
Outlet Gas

Atomizing mediums

- The selection of the atomizing jet medium is based mainly on the **reactivity** of the metal and the **cost** of the medium
- Air and water are inexpensive, but are reactive in nature
- Inert gases like Ni, Ar, He can be used but are expensive and hence have to be recycled
- Pumping of cold gas along with the atomizing jet => this will increase the solidification rate
- Synthetic oils are used instead of gas or water => this yields high cooling rate & lower oxygen content compared to water atomized powders
- **Oil atomization is suitable for high carbon steel, high speed steels, bearing steels, steel containing high quantities of carbide forming elements like Cr, Molybdenum**
- This method is not good for powders of low carbon steels

Process parameters

Effect of atomizing medium pressure

Increase in air pressure increases the fineness of powder up to a limit, after which no increase is seen

Molten metal temperature

As temperature increases, both surface tension and viscosity decrease; so available energy can efficiently disintegrate the metal stream producing fine powders than at lower temperature

Temperature effect on particle shape is dependent on particle temperature at the instant of formation and time interval between formation of the particle and its solidification

Temperature increase will reduce surface tension and hence formation of **spherical particle is minimal**
However **spherical particles can still be formed** if the disintegrated **particles remain as liquid for longer times.**

Orifice area: negligible effect

Molten metal properties:

- Iron and Cu powder => fine spherical size;
- Pb, Sn => irregular shape powder;
- Al powders => irregular shape even at high surface tension (oxidation effect)

Various Powder Production Methods

Sl. no.	Method	Purity	Particle characteristics Shape Mesh	Compresi- bility	Apparent density	Green strength
1	Atomization	Relatively good	Irregular to Coarse shots smooth, to 325 mesh rounded dense particles	Low to high	Generally high (spherical powder)	Generally low
2	Gaseous reduction of solids	Medium	Irregular, 100 mesh spongy and finer	Medium	Low to medium	High to medium
3	Gaseous reduction of	High	Irregular, 100 mesh spongy and finer	Medium	Low to medium	High
4	Reduction with carbon	Medium	Irregular, All meshes spongy from mesh size 8 downwards	Medium	Medium	Medium to high
5	Electrolytic deposition	High	Irregular, All mesh dendritic sizes	High (pure and ductile)	Medium to high	Medium
6	Carbonyl method	High	Spherical Usually in low micron ranges	Medium	Medium to high (spherical)	Low
7	Grinding	Medium	Flaky and All mesh dense sizes	Medium	Medium to low (flakes powders	Low (flaky particles 2D)
					A.D. low)	

PRODUCTION OF CERAMIC POWDERS

- Ceramics are generally made in powder form.
- The powders are then consolidated and sintered to give finished products.
- Conventional methods like casting, Forming cannot be used to process ceramics, because of their high-melting point and limited ductility.
- Several methods are used for the preparation of ceramic powders.
- (i) Mechanical methods
- (ii) Chemical methods.

Mechanical Methods

- In the mechanical methods, fine particles are produced from larger ones by mechanical comminution.
- Comminution can be achieved by crushing, grinding and milling.
- Powders of conventional ceramics are mainly prepared from naturally occurring raw materials by these methods.
- Mechanical size reduction is a cheap method and can be used for a wide variety of materials and is suitable for bulk production.
- **The limitations are relatively lower purity, limited chemical homogeneity as well as large particle sizes.**

Chemical Methods

- Powders of advanced ceramics (silicon nitride, SiAlON) are generally prepared by chemical methods from purified raw materials or chemicals synthesized artificially.
- High purity, fine ceramic powders can be prepared from solid, liquid or gas phase reactions.
- Chemical methods used for the production of ceramics fall under the following categories:
  (i) Solid state reactions.
  (ii) Precipitation from solution, and
  (iii) Vapour phase reactions.

Solid State Reactions

- Chemical decomposition, also known as calcination, is a simple process in which a solid chemical (reactant) is heated to produce a product, also in solid form, and a gaseous bye-product.
- The starting materials employed are usually compounds such as carbonates, hydroxides, nitrates, oxalates, alkoxides and other metallic salts.
- **A typical example is the calcination of magnesium carbonate to produce fine magnesia and carbon dioxide gas.**
- The final product usually requires crushing.
- The advantages of this method are finer particle size and relatively low-processing temperatures.
- Disadvantages include **lower purity, homogeneity limitation and the ability to process only simple oxides.**

- Solid state reaction is also used for the **production of complex oxides** like ferrites and titanate by the reaction between two chemicals in the solid state.
- The reaction between **barium carbonate and titania to produce barium titanate, a piezoelectric ceramic** is an example of this type of solid state reaction.
- Factors such as type of reactant (carbonate, or alkoxides), the initial size of the reactant particles as well as processing temperature and time affect the final powder characteristics.

Synthesis of Powders from Solution

- **Sol-gel technique**
- Sol-gel processing has attracted much interest both for the preparation of special powders and for forming thin coatings and cast or extruded shapes.
- In sol-gel processing, colloidal particles or molecules in a suspension, known as a sol, are mixed with a hydrolyzing agent, which causes them to join together into a continuous network, called a gel.
- Polymerization greatly restricts chemical diffusion and segregation.
- The gel is then dried, calcined and crushed to form a powder.

Preparation of silica gel from Tetraethyl orthosilicate

-

$$(OC_2H_5)_4Si + H_2O \xrightarrow{\text{hydrolysis}} (OC_2H_5)_3SiOH + C_2H_5OH$$

$$(OC_2H_5)_3SiOH + OH \xrightarrow{\text{Dehydration}} (OC_2H_5)_3SiO + H_2O$$

$$(OC_2H_5)_3SiO + (OC_2H_5)_3SiOH \xrightarrow{\text{Gelling}} (OC_2H_5)_3SiOSi(H_5C_2O)_3 + OH$$

On heating and calcination amorphous silica particles are obtained

Sol-gel method is used widely for the production of fine ceramic powders like yttria, titania and zirconia having high purity.

Evaporation of solution

- Evaporation, also called spray drying involves the spraying of a metallic salt solution in the form of fine droplets approximately 10–100 μm in diameter.
- Spraying is done in a heated chamber, which allows rapid evaporation.
- The powder agglomerates are then collected using suitable equipment.
- Important process parameters to be controlled include solution concentration, chamber temperature and gas flow rate.
- The primary particle size in the agglomerates can be very fine (0.1 μm or less).
- **In the freeze drying process**, the solution is sprayed by an atomizer, into a very cold medium such as liquid nitrogen.
- The resulting droplets are rapidly frozen before they have time to coalesce, removed and subjected to vacuum.
- This results in the removal of liquid nitrogen, leaving a very fine powder.
- Spray drying can also be used for metals.

ALLOY POWDER DESIGNATION

- Manufacturers of metal powders generally use trade names for their powders since many of the powders are proprietary in nature.
- Generic specification relating to elemental or alloy powders are available from two standard organizations, namely the Metal Powder Industries Federation (MPIF) Princeton, New Jersey, USA and the American Society for Testing and Materials (ASTM) Ohio, USA.
  - MPIF has provided a metal powder designation system (MPIF Standard 3.5), according to which, metal and alloy powders are given a prefix and a four-digit code.
  - The prefix will refer to the metals present while the four-digit code will give the percentage of the alloying elements present.
  - The minimum yield strength is referred to by another two-digit code at the end.

Table 2.2 MPIF Alloy Powder Designation System

Prefix	*Element referred*
A	Aluminium
C	Copper
D	Molybdenum
F	Iron
G	Graphite
M	Manganese
N	Nickel
P	Lead
S	Silicon
SS	Stainless steel
T	Tin
Z	Zinc

- Alloy with designation CFG-xxyy-zz refers to copper-iron-graphite powder, where xx refers to the % of iron and yy refers to the % of graphite.
- The letters zz refers to the minimum yield strength in ksi.

- For example, the alloy F-0000-10 refers to pure iron powder with a nominal strength of 10 ksi.
- Both the MPIF and ASTM standards are widely accepted by powder as well as component manufacturers.